Politecnico di Torino

AMML

Advanced Machine Learning

A.A. 2023/2024

Tatiana Tommasi

A short survey

Do you know what is Principal Component Analysis (PCA)?

Do you know what is Support Vector Machines (SVM)?

Have you ever trained an SVM classification model?

Do you know what is cross validation?

Do you have experience with python coding?

Have you ever trained a CNN classification model?

Basic Information

Teacher: Tatiana Tommasi
http://www.tatianatommasi.com/
tatiana.tommasi@polito.it

Support from:
Antonio Alliegro, Paolo Rabino, Leonardo Iurada, Claudia Cuttano...
Prof. Barbara Caputo, Giuseppe Averta, Carlo Masone, Raffaello Camoriano
Visual and Multimodal Applied Learning Laboratory (Vandal Lab)
http://vandal.polito.it/

Code: 01URWOV
Master of science in Computer Engineering (Ingegneria Informatica)
Credits: 6CFU

Basic Information

When/Where: Tuesday 16:00-19:00, room 3s
Friday 11:30-13:00, room 1s

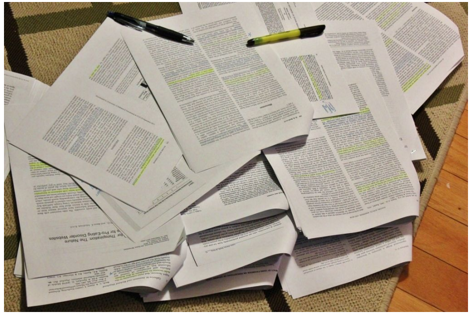
How: in presence + online (rare pre-recorded videos) + discussion sessions
Exercises: in presence
everything will be recorded with exception of the exercise sessions.

Slack channel:
https://join.slack.com/t/aml2023/signup
you should access with your @studenti.polito.it email account
It will be useful mainly to interact with the project mentors

Useful Material

Deep Learning. Ian Goodfellow, Yoshua Bengio, Aaron Courville, https://www.deeplearningbook.org/
The little book of Deep Learning https://fleuret.org/francois/lbdl.html
Artificial Intelligence A Modern Approach. Stuart J. Russel and Peter Norvig
An Introduction to Probability Theory and its applications. William Feller
Machine Learning: a probabilistic perspective. K. P. Murphy.
Understanding Machine Learning: From Theory to Algorithms. Shai Shalev-Shwartz and Shai Ben-David

CVPR
NeurIPS
ECCV
ICCV
ICLR
…
arXiv



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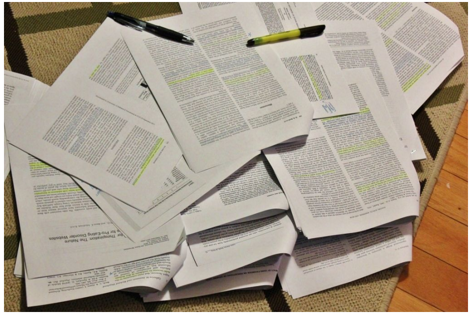

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Specific references will be given throughout the course in the form of **book chapters** and **scientific articles**.

Courses:
online - Convolutional Neural Networks for Visual Recognition - Fei Fei Li - http://cs231n.stanford.edu/
online - Machine Learning course on Coursera - Andrew Ng - https://www.coursera.org/

polito - Data Spaces - 01RLPOV, 01RLPNG, Prof. Vaccarino
polito - Machine Learning and Pattern Recognition - 01URTOV, Prof. Cumani

Pre-requirement: Python basic elements
Online tutorial for Python: https://docs.python.org/3/tutorial/
Online tutorial for Pytorch: https://pytorch.org/tutorials/

Topics and Objective

- AI, ML basic definitions
- basic probability
- loss, risk, prediction rules
- ~~shallow supervised/unsupervised learning~~
- deep learning (supervised, unsupervised, recurrent, style transfer)
- **computer vision applications**

Topics and Objective

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+ 3d geometry
+ natural language processing

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ellis
European Laboratory for Learning and Intelligent Systems

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Alicante	Amsterdam	Berlin	Cambridge
Copenhagen	Darmstadt	Delft	Edinburgh
Lausanne (EPFL)	Zürich (ETH)	Freiburg	Genoa
Haifa (Technion)	Heidelberg	Helsinki	Jena
Vienna (IST Austria)	Leuven	Linz	Lisbon
London (UCL)	Madrid	Manchester	Milan
Modena (Unimore)	Munich	Nijmegen	Oxford
Paris	Prague	Saarbrücken	Stuttgart
Tel Aviv	Tübingen	$\boxed{\text{Turin}}$	

Unit Director

Tatiana Tommasi
Scholar

Unit Members

Barbara Caputo
Fellow

Diego Valsesia
Member

Enrico Magli
Fellow

Giulia Fracastoro
Member

Giuseppe Averta
Member

Giuseppe Rizzo
Member

Raffaello Camoriano
Member

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Google Scholar

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Member

Course Aim

- You will get to know the basics of ML & DL
- You will learn how to read & understand a paper (code included)
- You will learn how to design & implement your own ML/DL algorithm for a given problem
- You will learn how to explain what you know about ML/DL to others

Course Aim

- You will get to know the basics of ML & DL

  ~~mix of~~ in-person~~, online live and recorded lectures~~

- You will learn how to read & understand a paper (code included)

  individual/small group interactive work guided by me and by teaching assistants

- You will learn how to design & implement your own ML/DL algorithm for a given problem

  work in groups mainly guided by the teaching assistants

- You will learn how to explain what you know about ML/DL to others

  presentation in front of the class / at the exam

Exam Modality

Paper Presentation + Project + Oral Exam

Exam Modality

Paper Presentation + Project + Oral Exam

By mid-November (it might vary a bit) we will have covered basics DL

Then the lectures will be dedicated to presentation of seminal papers of the last 3-7 years (roughly) on various DL topics. The topics will be of interest for all projects.

You will present papers in groups: **3 students, 2 papers**

Papers will be assigned by me in random order.
You will know the title of your paper 2 weeks before your presentation. There will be some time to adjust the calendar.

You will have **10 minutes sharp** for the presentation.
You will be judged on your comprehension of the papers and on your ability to explain it.

Minimum individual score: 6
Maximum individual score : 10

Exam Modality

Paper Presentation + Project + Oral Exam

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HOW
There will be a lecture dedicate to how to present a paper.

WHY????
Because working with ML/DL means being a life-long-learner…

Exam Modality

Paper Presentation + Project + Oral Exam

At the beginning of November one lecture will be dedicated to possible project research-areas.

There will be >4 proposals, for each proposal there will be some mandatory work plus suggestions on how to extend it.

You will work in teams of 3 people.

Project Proposal Submission: Before Christmas break (more details on date / submission modality will come), you will indicate: (1) your team (2) the project you chose (3) your planned extension

Final report due: ten days before the oral exam
The report will be a pdf of max 8 pages using a provided template

You can keep your project till the last exam session before the next course beginning (1 year)

Minimum group score: 6
Maximum group score : 10

Exam Modality

Paper Presentation + Project + Oral Exam

You will make a group presentation (about 15 mins) of your project and a Q&A session will follow

Minimum individual score : 6
Maximum individual score : 10

Important Note (1) : This is not a trial-error exam
Important Note (2): the code of your project will be checked - don’t cheat

Questions: Paper Presentation

Paper Presentation:

- Can I chose the paper?
  No

- I cannot make the presentation on my assigned date because ...
  You can ask a colleague for a switch: in this case I have to receive an email in advance explaining why the switch happens and who will present the paper. No impact on the score either way.

- I do not make the presentation on the assigned day & I don’t make the switch...
  You get 0 points

- Will I get guidance/help in preparing the presentation?
  Yes I will make one lecture on how to read a scientific paper & one lecture on how to present a scientific paper/work. You will not get individual help or support to try out the presentation.

Questions: Project

- Can I propose my own project?
  No

- Can I do the project alone/in two?
  No. If you don’t know anyone we’ll match you

- What computing facilities I can use for the project?
  Google Colab

- How will I chose the project?
  There will be >4 projects, each led by a different teaching assistant.
  In November each project will be presented with a lecture & live Q&A to help you choosing

- I am not proficient in python, will I be able to make the project?
  It will be tough but you can exploit a lot of online material to learn python…if you want to make a living in ML you better learn it!

- Will I get guidance/help in working on the project?
  Yes you will be supported by the teaching assistant both with meetings and on slack.

Questions: Oral

- Can I make the oral exam in a different session than the rest of my group?
  No

- Is the oral exam in English?
  Yes

- What will be asked during the exam?
  Everything related to the project, from implementation details to theory related to the algorithms used in the project. There will be also questions on any topic explained during the lectures.

- Why are we doing this project???? 😭
  This is the main question I'm going to ask you during the exam…

- Can I reject the final vote and sit again the exam?
  Yes. For the next exam session you will need to search for another team on a different topic (i.e. you will have to work on a different project). Please come to sit at the exam when you are prepared.

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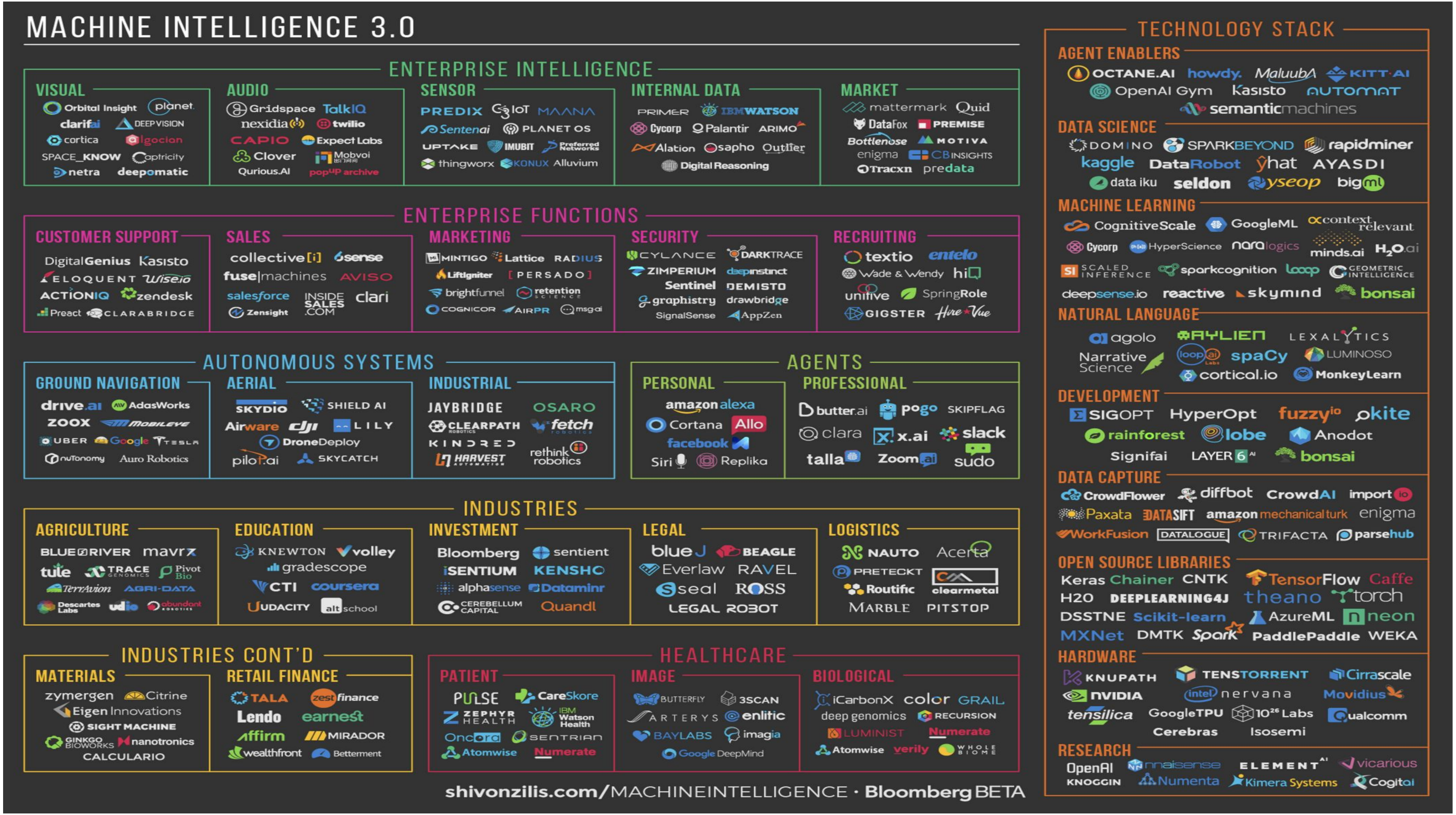
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AUTONOMOUS SYSTEMS
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drive.ai AdasWorks zoox MOBILEYE UBER Google TESLA nuTonomy Auro Robotics
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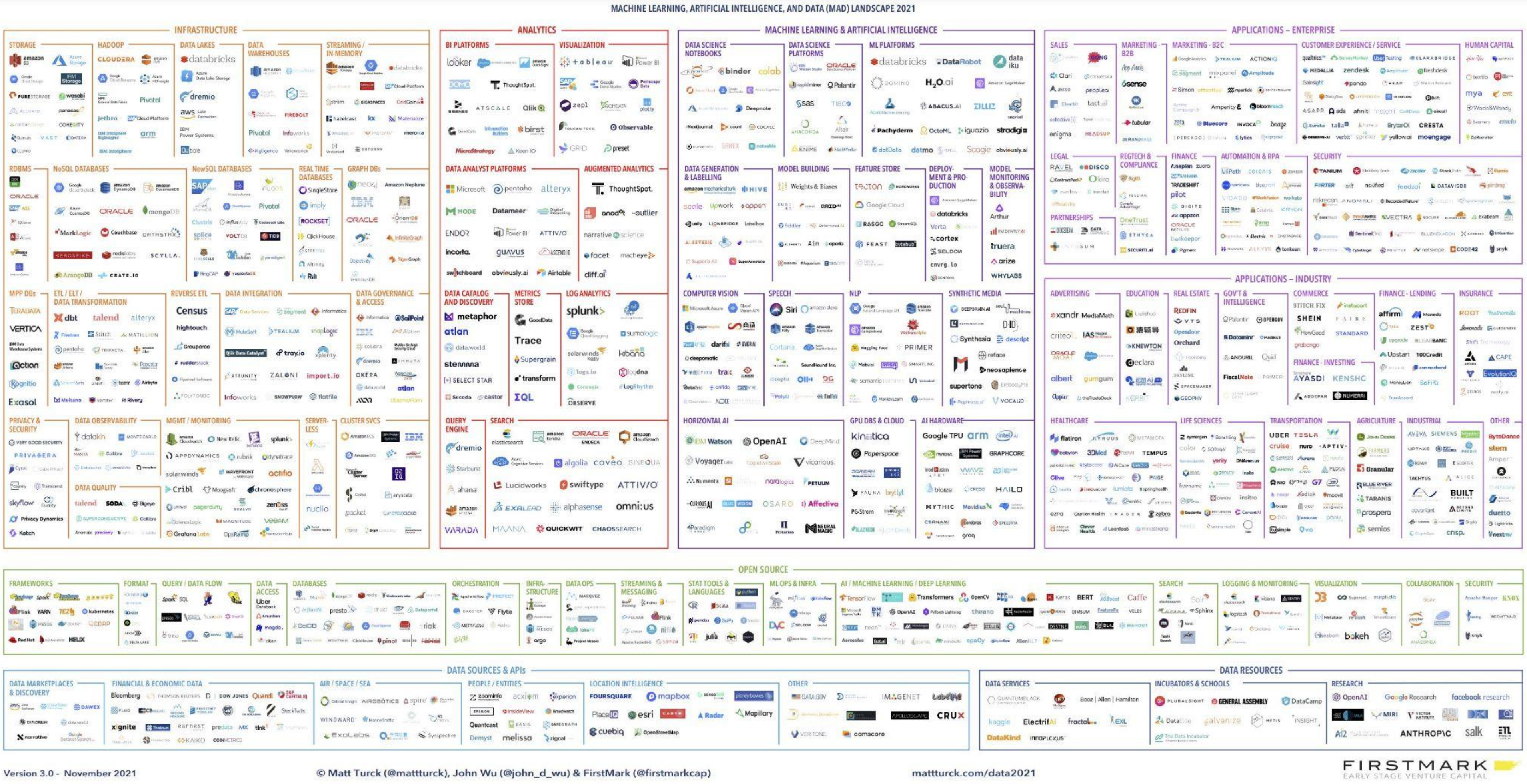
RESEARCH
OpenAI nnaisense ELEMENT AI vicarious KNOGGIN Numenta Kimera Systems Cogitai


MACHINE LEARNING, ARTIFICIAL INTELLIGENCE, AND DATA (MAD) LANDSCAPE 2021
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STORAGE
HADOOP
DATA LAKES
DATA WAREHOUSES
STREAMING / IN-MEMORY
RDBMS
NoSQL DATABASES
NewSQL DATABASES
REAL TIME DATABASES
GRAPH DBs
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REVERSE ETL
DATA INTEGRATION
DATA GOVERNANCE & ACCESS
PRIVACY & SECURITY
DATA OBSERVABILITY
DATA QUALITY
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BI PLATFORMS
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SYNTHETIC MEDIA
HORIZONTAL AI
GPU DBS & CLOUD
AI HARDWARE
APPLICATIONS – ENTERPRISE
SALES
MARKETING - B2B
MARKETING - B2C
CUSTOMER EXPERIENCE / SERVICE
HUMAN CAPITAL
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REGTECH & COMPLIANCE
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SECURITY
PARTNERSHIPS
APPLICATIONS – INDUSTRY
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EDUCATION
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GOV'T & INTELLIGENCE
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FINANCE - LENDING
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AIR / SPACE / SEA
PEOPLE / ENTITIES
LOCATION INTELLIGENCE
OTHER
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RESEARCH
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Why Are you attending this course?

Learn about basic tools that you will need for:

- **your M. Sc./PhD Thesis**
- **your Start Up**
- **Wall Street**
- **Google, Facebook, Amazon, Snapchat, ...**
- **Huawei, Samsung, IBM, Siemens, Microsoft, Apple,...**
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Caveat…

- NOT: learn to use deep learning as a black box
- YES: understand models and architectures

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This is not an easy course, but results speak for themselves!

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Top 10 jobs involving AI skills

Top jobs seeking machine learning or artificial intelligence skills

Rank	Job title	% of postings containing AI or machine learning	Rank	Job title	% of postings containing AI or machine learning
1.	Machine learning engineer	75.0%	6.	Algorithm developer	46.9%
2.	Deep learning engineer	60.9%	7.	Junior data scientist	45.7%
3.	Senior data scientist	58.1%	8.	Developer consultant	44.5%
4.	Computer vision engineer	55.2%	9.	Director of data science	41.5%
5.	Data scientist	52.1%	10.	Lead data scientist	32.7%

Source: Indeed

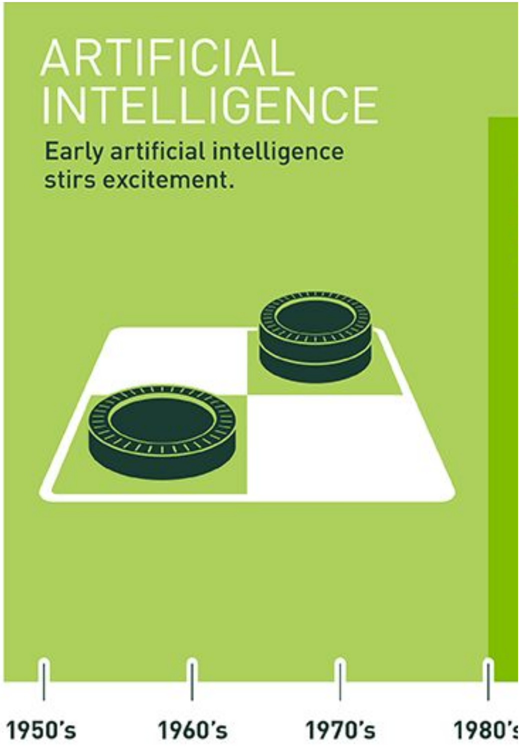
indeed

AI job titles with the highest salaries

Rank	Job title	Average salary
1.	Machine learning engineer	$142,858.57
2.	Data scientist	$126,927.41
3.	Computer vision engineer	$126,399.81
4.	Data warehouse architect	$126,008.25
5.	Algorithm engineer	$109,313.51

AN ML

What is AI?



https://developer.nvidia.com/deep-learning

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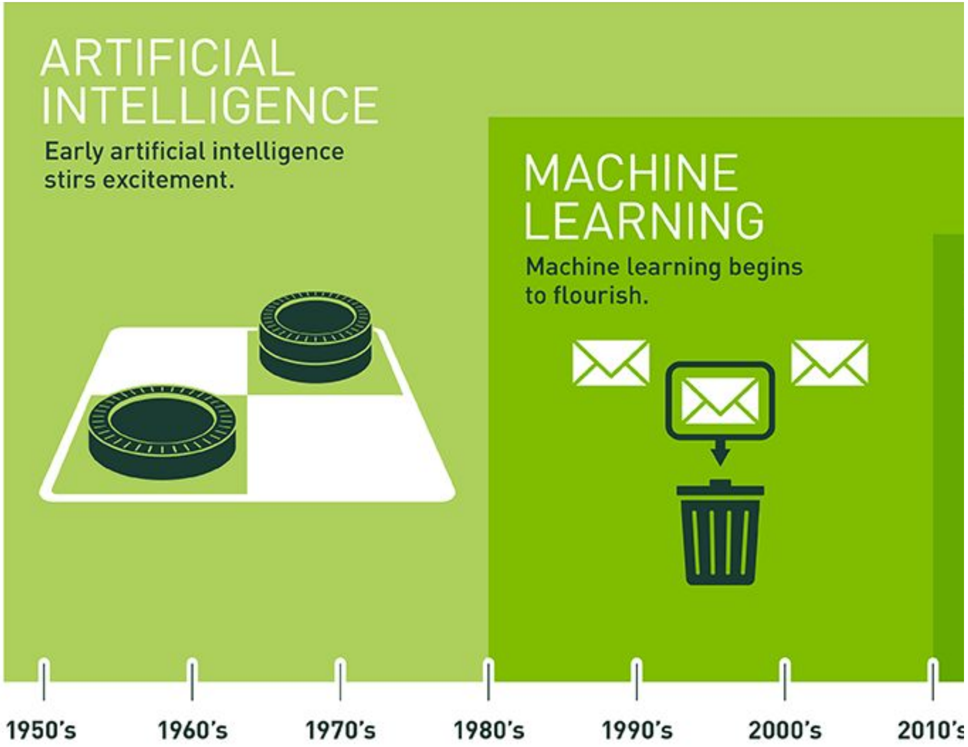

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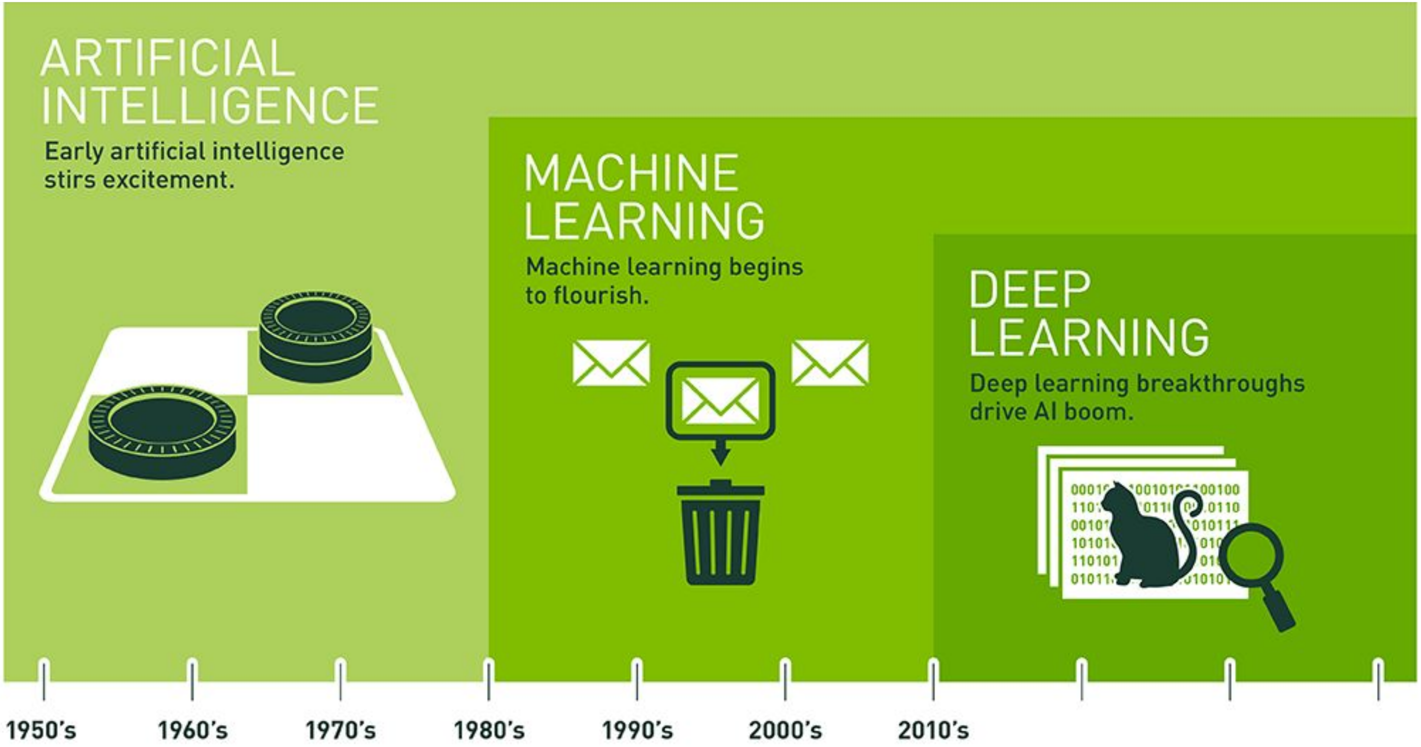

Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

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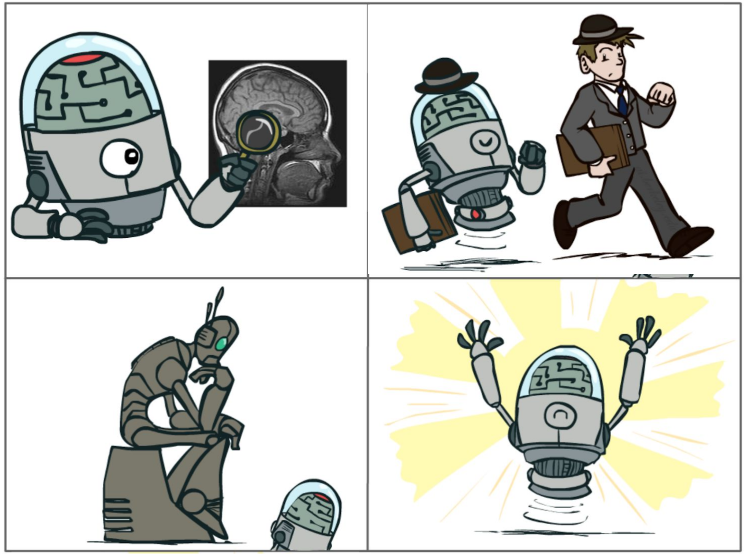
What is AI?

Thinking humanly

Acting humanly



Thinking rationally

Acting rationally

http://www-inst.eecs.berkeley.edu/~cs188/fa19/
http://slazebni.cs.illinois.edu/fall17/

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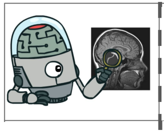
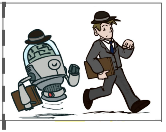
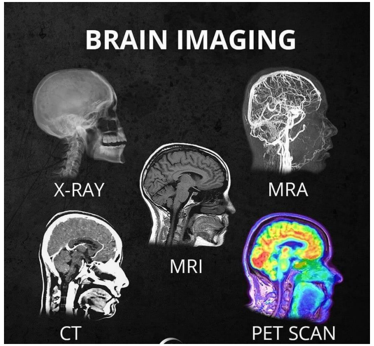




Acting humanly

Make computers think like a human
How does a human think?

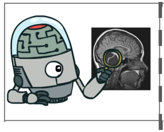
- brain imaging - neuroscience
- psychological experiments - cognitive science

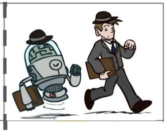


Alison Gopnik - developmental scientist

What is AI?

Thinking humanly





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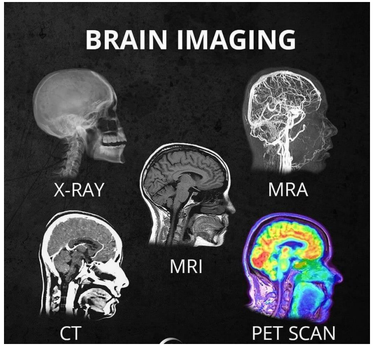
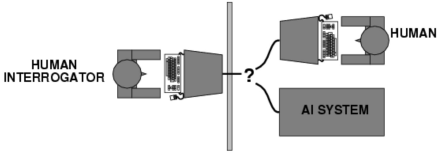
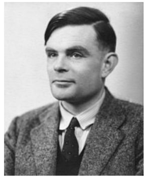


Alison Gopnik - developmental scientist

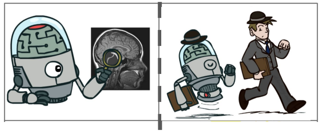
The Turing Test Approach





What is AI?

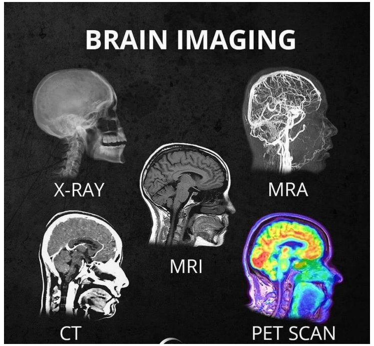
Thinking humanly



Acting humanly

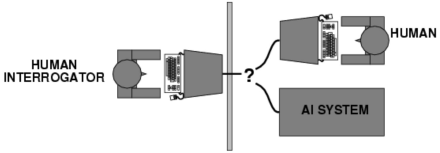
Make computers think like a human
How does a human think?

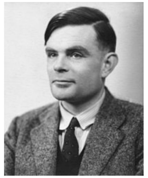
- brain imaging - neuroscience
- psychological experiments - cognitive science



Alison Gopnik - developmental scientist

The Turing Test Approach





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- **knowledge representation** - store knowledge
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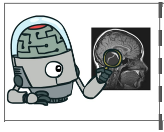

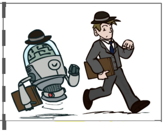

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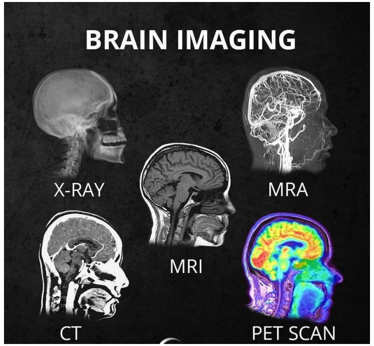

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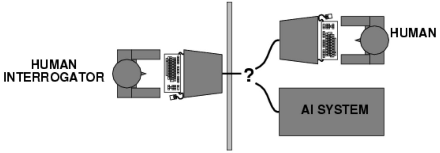

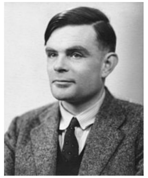

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Aeronautical engineering *"goal of the field is making machines that flies so exactly like pigeons that they can fool other pigeons"*

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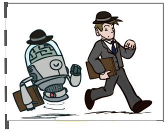
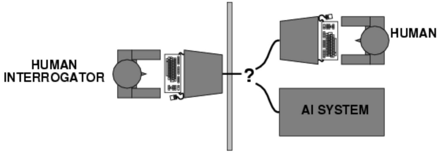


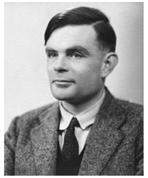
Acting humanly

Issues with this test:

- Variability in protocols, judges
- Success depends on deception!
- Chatbots can do well using “cheap tricks” (Eliza 1966)

The Turing Test Approach





Winograd schema: Multiple choice questions that can be easily answered by people but cannot be answered by computers using “cheap tricks”

The sack of potatoes had been placed below the bag of flour, so it had to be moved first. What had to be moved first?
• The sack of potatoes
• The bag of flour

https://ptchallenge-workshop.github.io/

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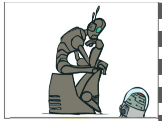
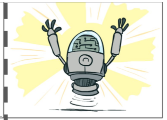




Acting rationally

Idealized or “right” way of thinking

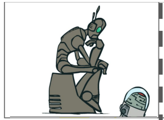
Logic: patterns of argument that always yield correct conclusions when supplied with correct premises

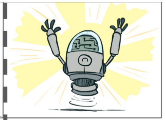
Logicist approach to AI: describe problem in formal logical notation and apply general deduction procedures to solve it

Issues:
• Describing real-world problems and knowledge in logical notation
• Dealing with uncertainty
• Computational complexity of finding the solution starting from a list of facts

What is AI?

Thinking rationally





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A rational agent *acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.*

- Goals are expressed in terms of the utility of outcomes (quantitative measure)
- Naturally accommodates uncertainty
- Rationality only concerns what decisions are made (no cognitive process behind)
- Being rational means maximizing your expected utility

In practice, utility optimization is subject to the agent's computational constraints (bounded rationality or bounded optimality)

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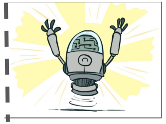
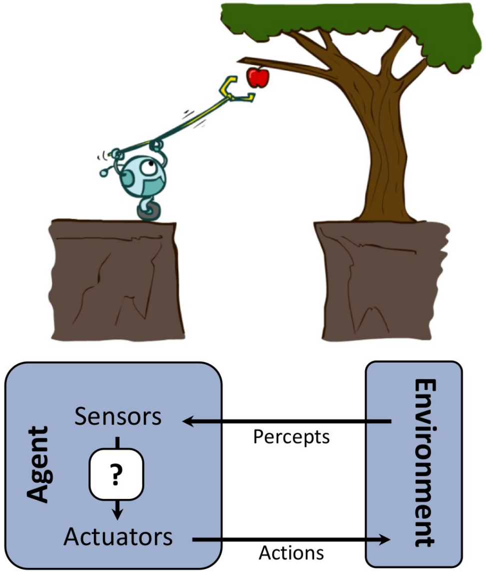


Acting rationally

AI = Computational Rationality



http://www-inst.eecs.berkeley.edu/~cs188/fa19/

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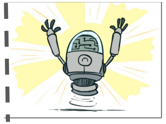

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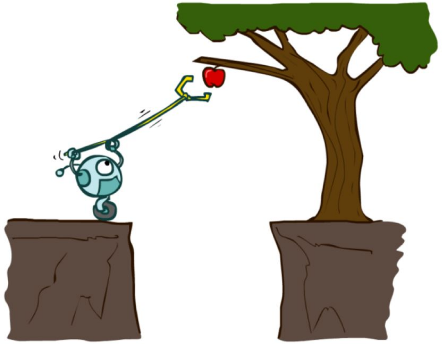

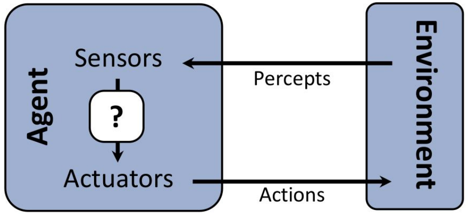

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All the skills needed for the Turing Test allow an agent to act rationally

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enable the agent to reach good decisions

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generate comprehensible sentences and get by in a complex society

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improve ability to generate effective behavior

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interact beyond language communication

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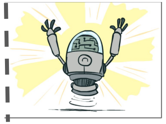

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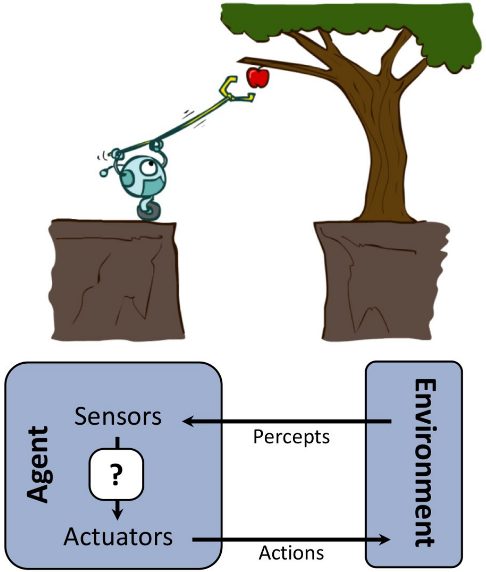

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AN ML

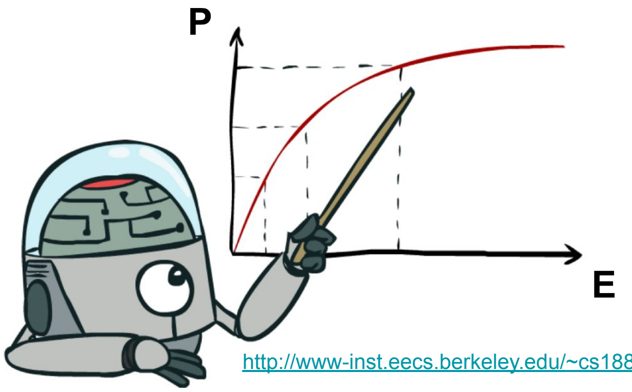
Machine Learning

How do we define “Learning”?

1. experience E (data)
2. task T
3. performance measure P

An agent learns when its performance at task T, measure by P, improves with experience E.

We need to get a computer do well on a task without manually building its competence.



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Machine Learning

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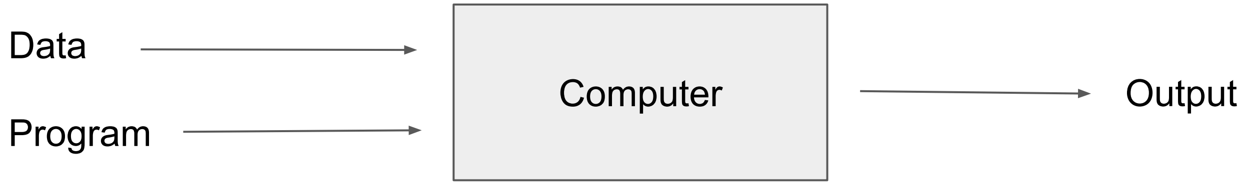
T : Playing checkers
E : Playing practice games against itself
P : Percentage of games won against an arbitrary opponent

T : Recognizing hand-written words
E : Database of human-labeled images of handwritten words
P : Percentage of words correctly classified

T : Driving on four-lane highways using vision sensors
E : A sequence of images and steering commands recorded while observing a human driver.
P : Average distance traveled before a human-judged error

T : Categorize email messages as spam or legitimate
E : Database of emails, some with human-given labels
P : Percentage of email messages correctly classified

Programming with Data



We want adaptive, robust and fault tolerant systems
Rule based implementation (often)

- is difficult (for programmers)
- is brittle (can miss many edge cases)
- becomes a nightmare to maintain explicitly
- does not work too well

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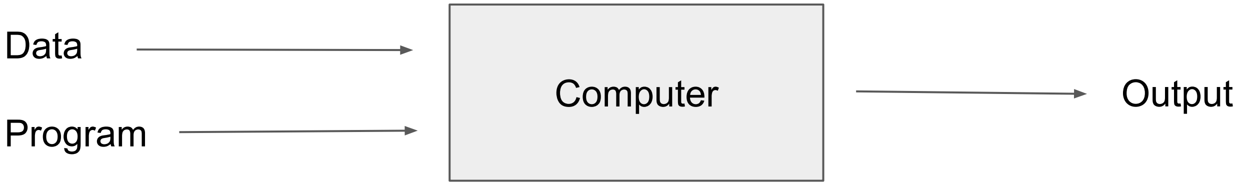

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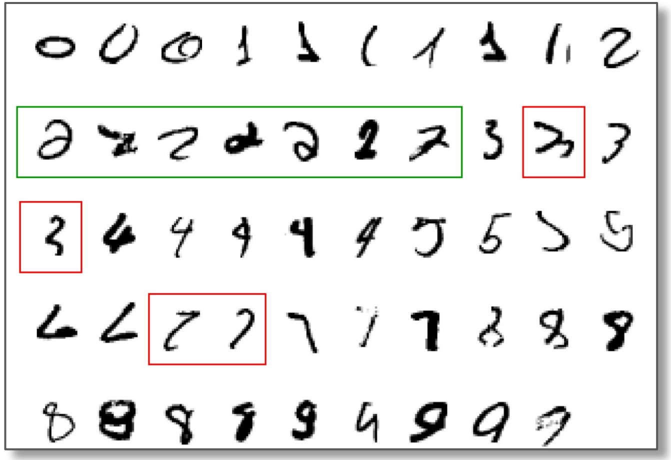
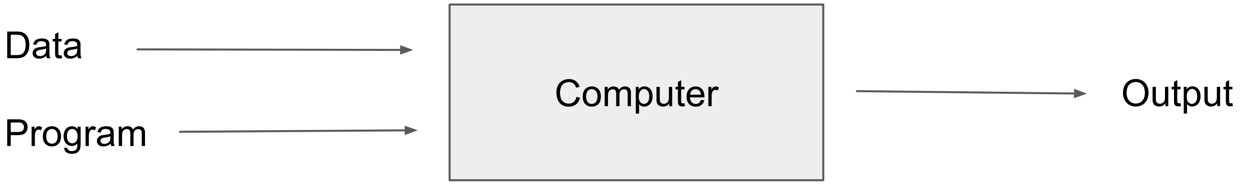


figure credit: G. Hinton

Programming with Data



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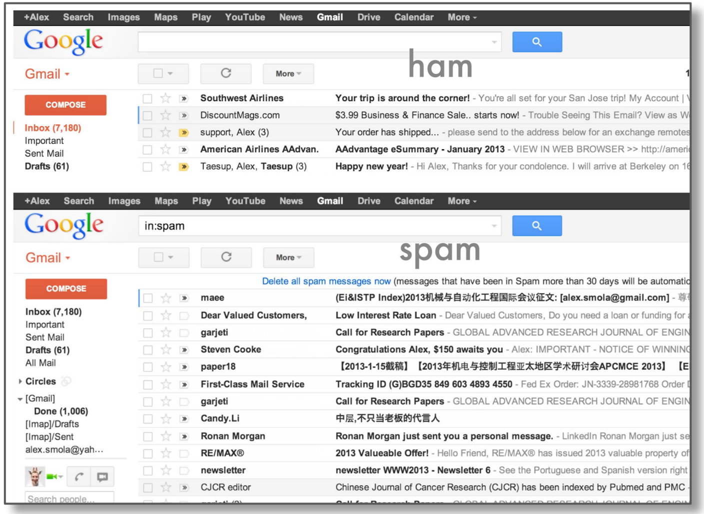

figure credit: A. Smola

Programming with Data

Data →
Program →
Computer
→ Output

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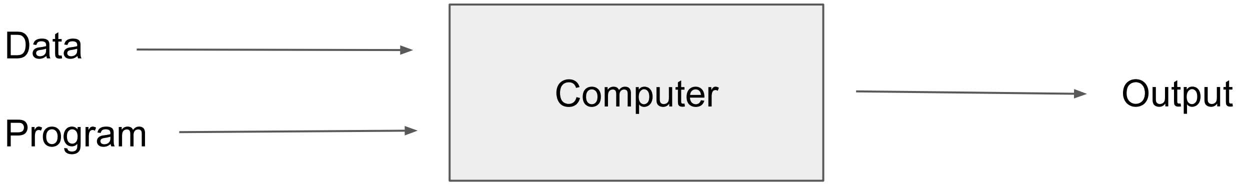

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➢ It is easier to obtain examples of what we want: **IF x THEN DO y**
➢ collect many pairs (x_i, y_i)
➢ estimate a function f such that $f(x_i) = y_i$ (supervised learning)
➢ detect patterns in data (unsupervised learning)

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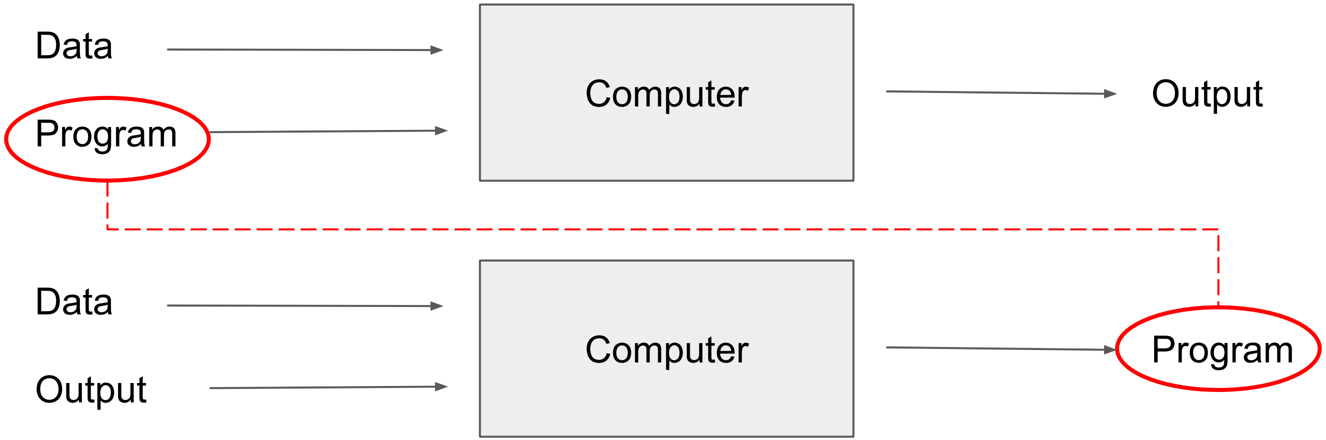

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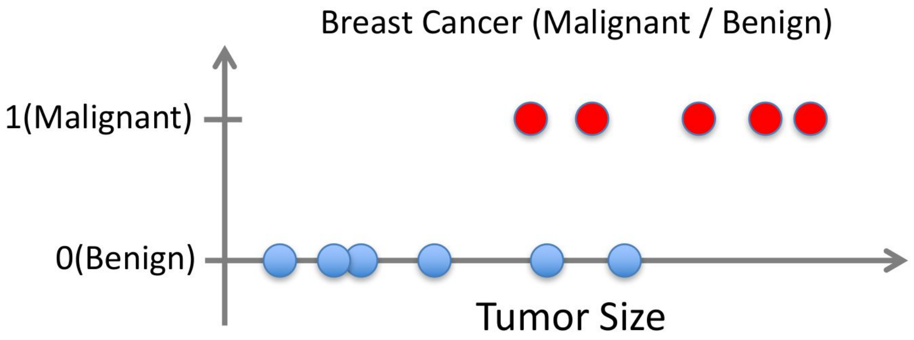
AI ML

Supervised Learning: Classification

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is categorical == classification

Binary Classification
y={-1,1}



slide credit: A.Ng

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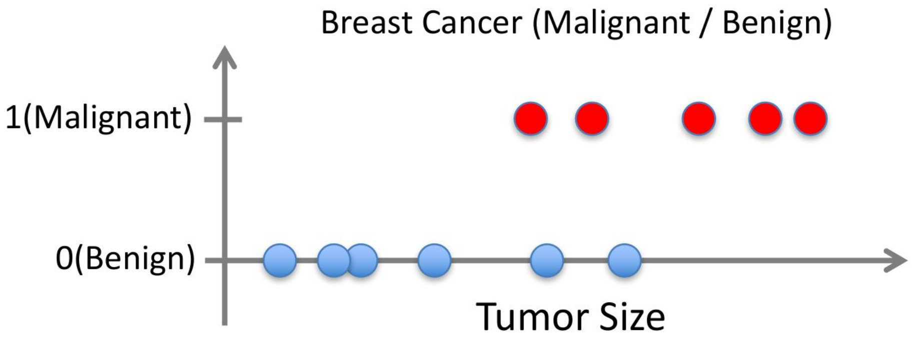

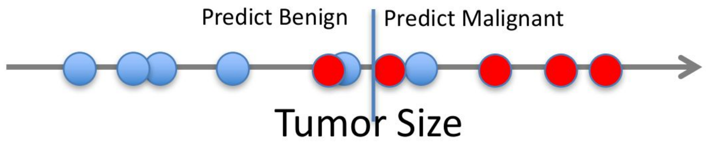

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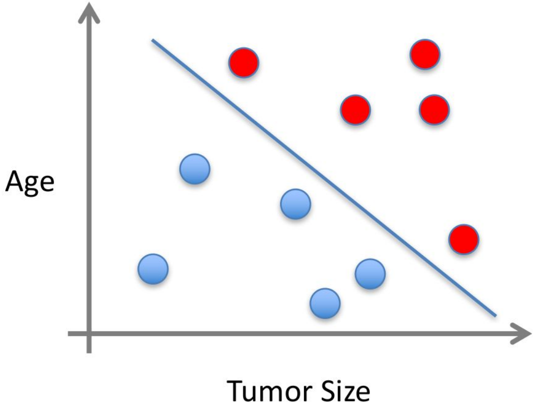
Supervised Learning: Classification

- x can be multi-dimensional
  – Each dimension corresponds to an attribute

Binary Classification
y={-1,1}

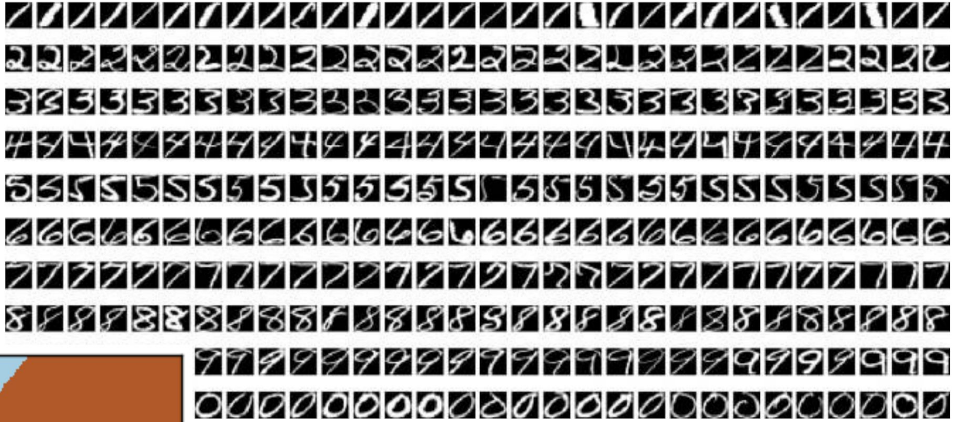


- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape

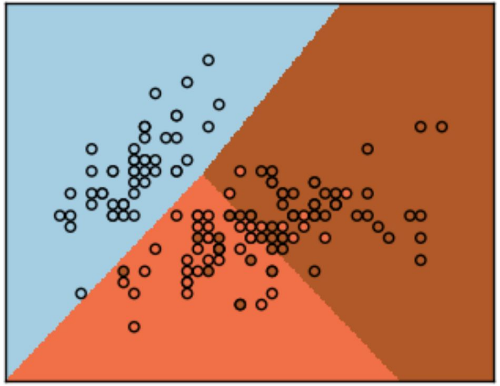
...

Supervised Learning: Classification

Multiclass Classification
y={1,...,k}



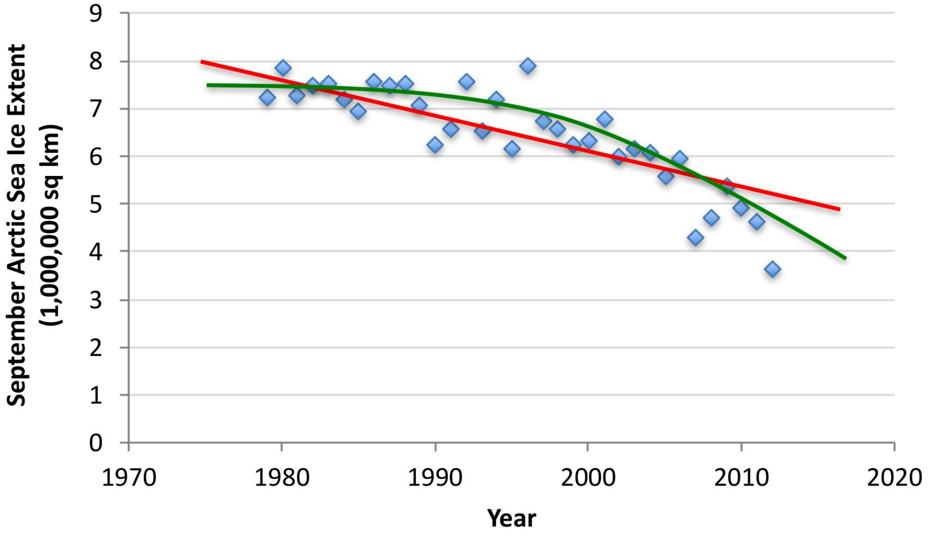
map image x to digit y



Supervised Learning: Regression

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is real-valued == regression

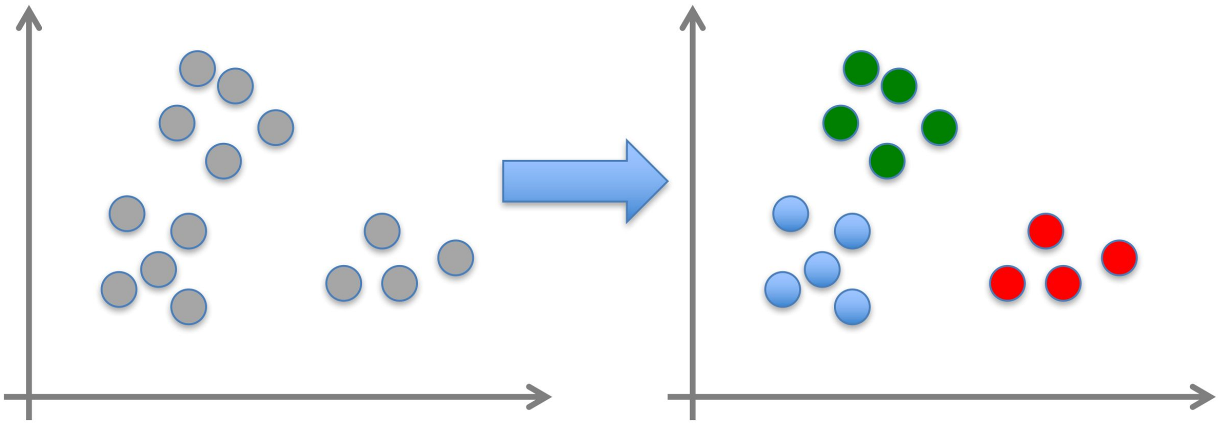
red: linear
green: non linear



slide credit: E Eaton

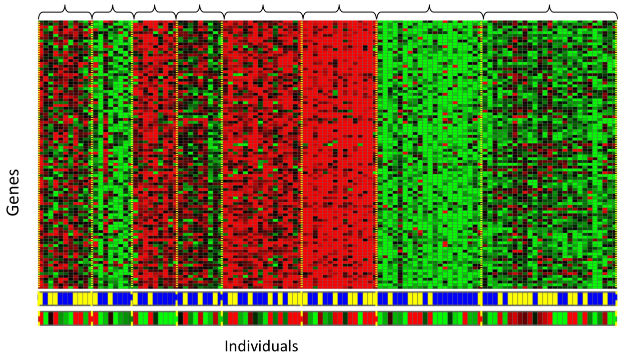
Unsupervised Learning

- Given $x_1, x_2, \ldots, x_n$ (without labels)
- Output hidden structure behind the x's
  - E.g., clustering

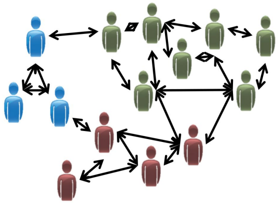


Unsupervised Learning

Genomics application: group individuals by genetic similarity

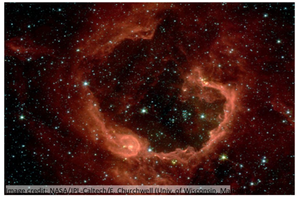


Organize computing clusters

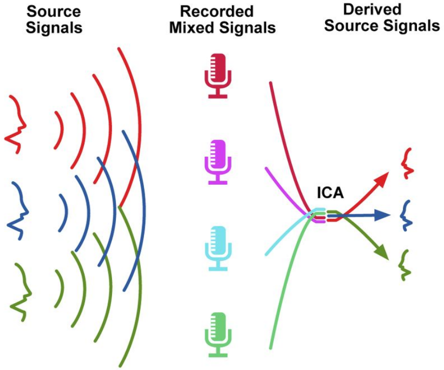


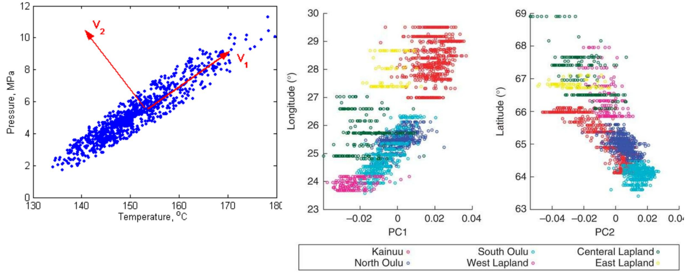
Social network analysis

Market segmentation



Astronomical data analysis



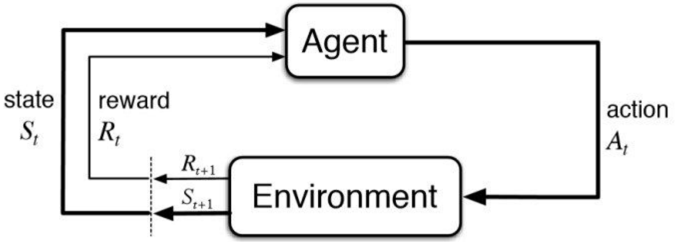


Variance component model to account for sample structure in genome-wide association studies, Nature Genetics 2010

figures credit: E Eaton

Many more Learnings...

- Reinforcement Learning



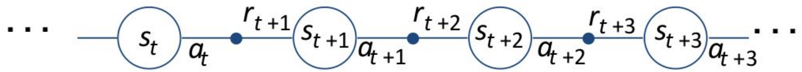
Agent and environment interact at discrete time steps : $t = 0, 1, 2, \ldots$

Agent observes state at step t: $s_t \in S$

produces action at step t: $a_t \in A(s_t)$

gets resulting reward : $r_{t+1} \in \Re$

and resulting next state : s_{t+1}



figures credit: Sutton & Barto

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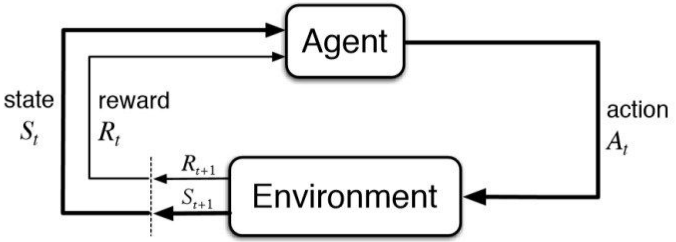

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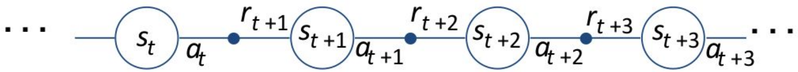

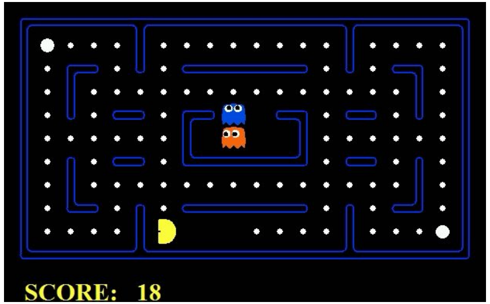

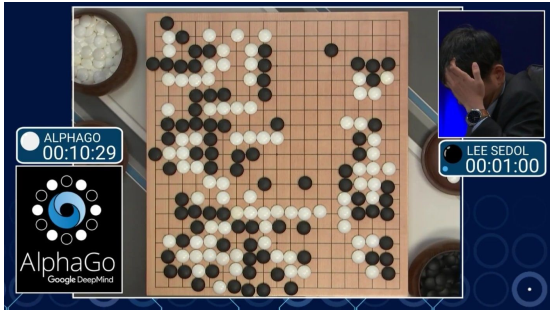

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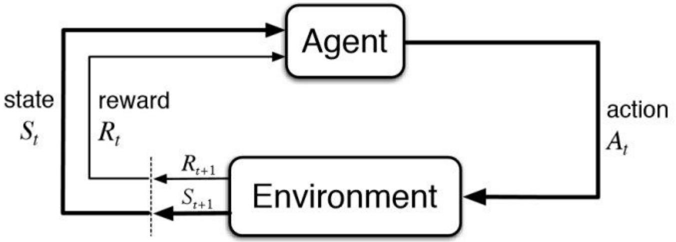


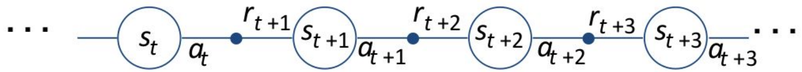
Agent and environment interact at discrete time steps : $t = 0, 1, 2, K$

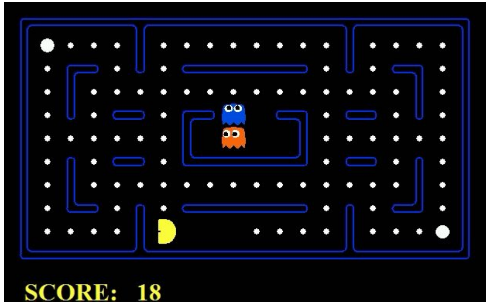
Agent observes state at step t: $s_t \in S$

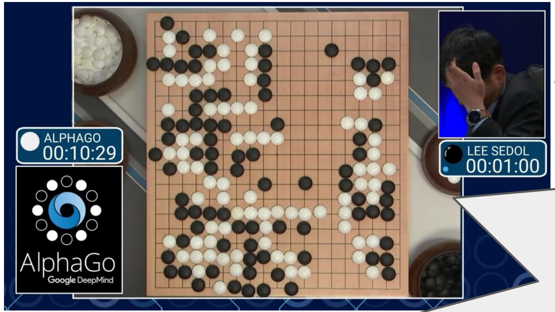
produces action at step t: $a_t \in A(s_t)$

gets resulting reward : $r_{t+1} \in \Re$

and resulting next state : s_{t+1}



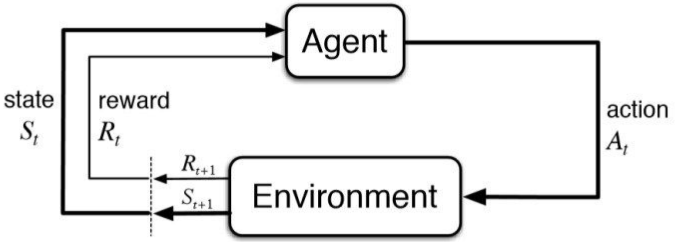




New Course
Robot Learning
01HFNOV

Many more Learnings...

- Reinforcement Learning



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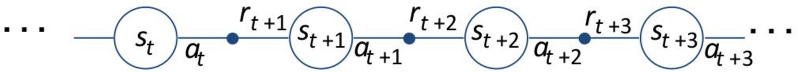
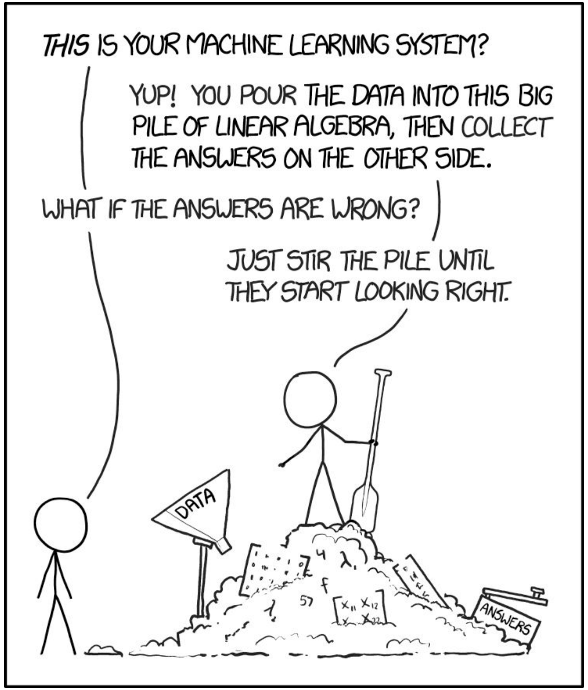


- Active Learning
- Online Learing
- Multi-task Learning
- Metric Learning
- Ensemble Learning
- Self-Supervised Learning
- Discriminative vs Generative Learning
- [...]

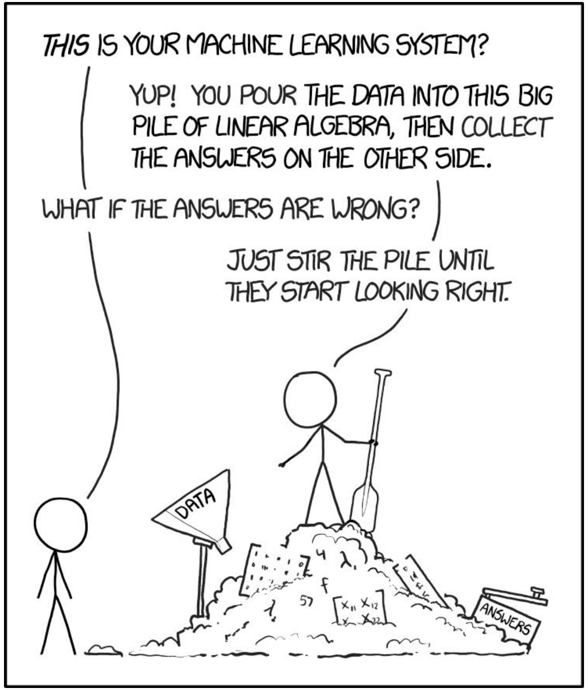
figures credit: Sutton & Barto

Who is a Machine Learner?



https://xkcd.com/1838/

Who is a Machine Learner?



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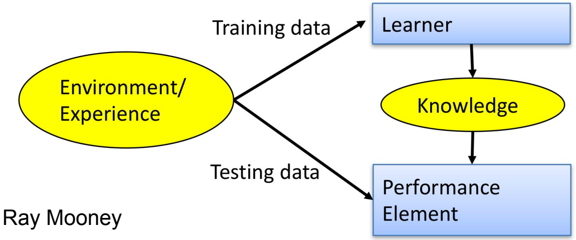
- Framing a Learning Problem
- Designing a Learning System
    - choose the training experience
    - choose exactly what is to be learned (target function)
    - choose a learning algorithm to infer the target function from the experience



Based on a slide of Ray Mooney

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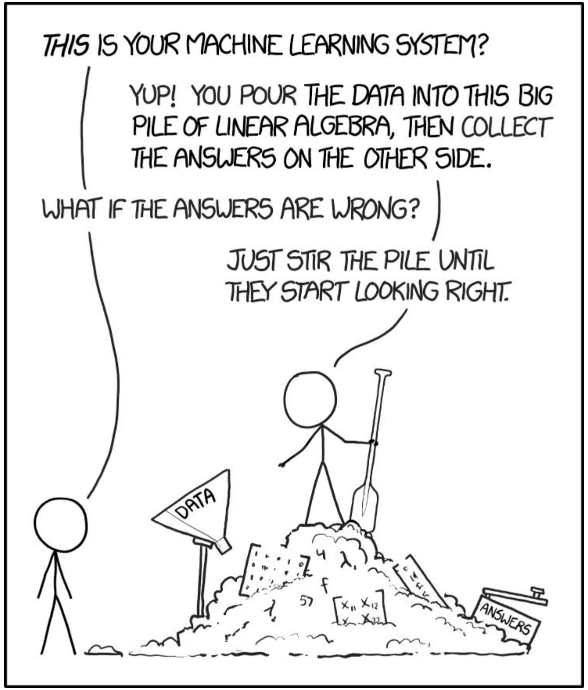

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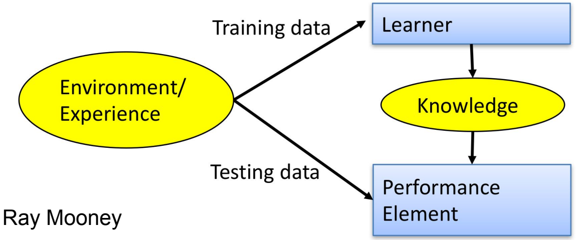
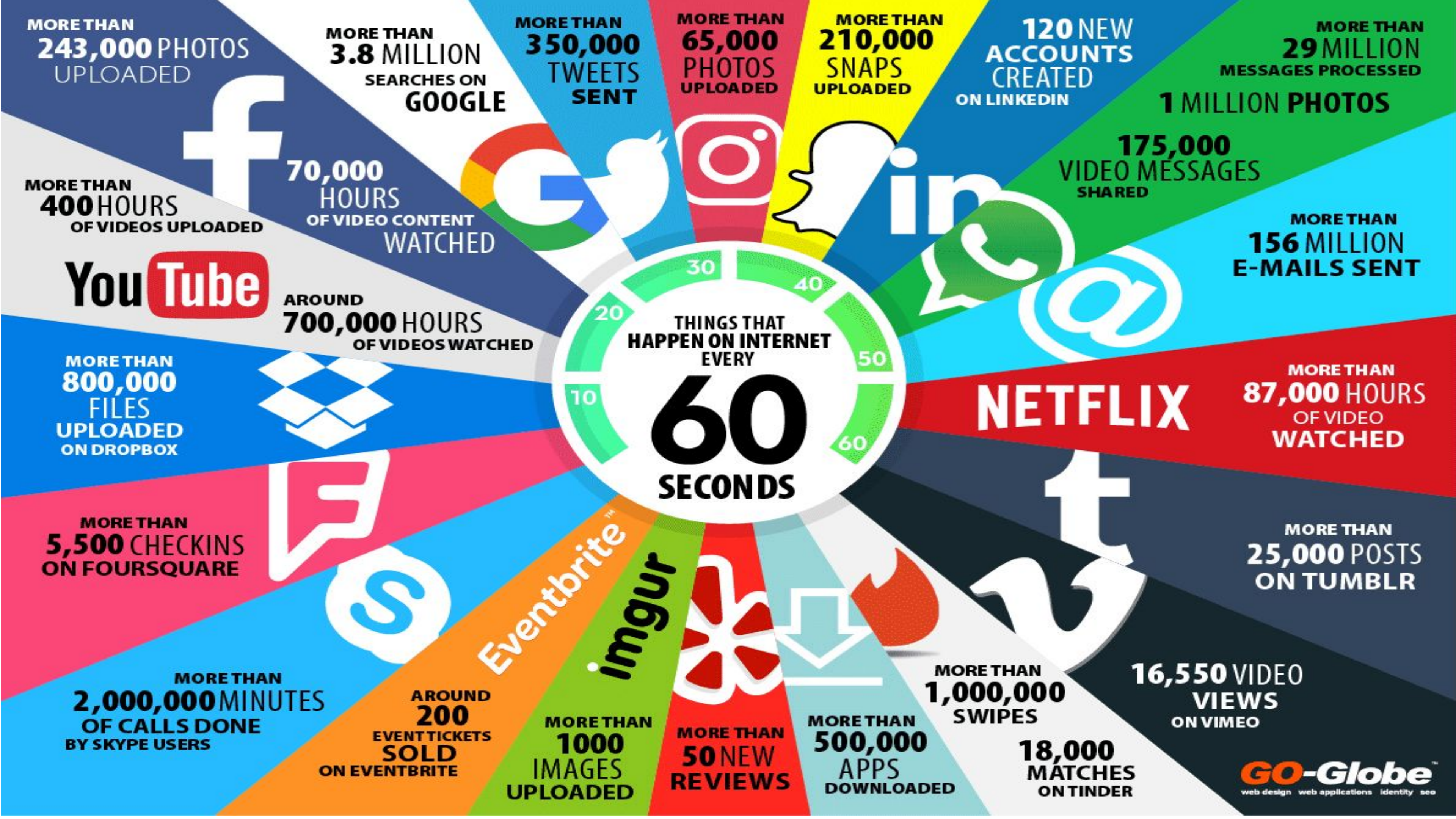


Based on a slide of Ray Mooney

Pseudo-Algorithm

1. Understand domain, prior knowledge, and goals
2. Data integration, selection, cleaning, pre-processing, etc.
3. Learn models
4. Interpret results
5. Consolidate and deploy discovered knowledge
6. Go back to 1



MORE THAN 243,000 PHOTOS UPLOADED
MORE THAN 3.8 MILLION SEARCHES ON GOOGLE
MORE THAN 350,000 TWEETS SENT
MORE THAN 65,000 PHOTOS UPLOADED
MORE THAN 210,000 SNAPS UPLOADED
120 NEW ACCOUNTS CREATED ON LINKEDIN
MORE THAN 29 MILLION MESSAGES PROCESSED
1 MILLION PHOTOS
175,000 VIDEO MESSAGES SHARED
70,000 HOURS OF VIDEO CONTENT WATCHED
MORE THAN 400 HOURS OF VIDEOS UPLOADED
YouTube
AROUND 700,000 HOURS OF VIDEOS WATCHED
MORE THAN 156 MILLION E-MAILS SENT
30
40
20
50
10
60
THINGS THAT HAPPEN ON INTERNET EVERY
60
SECONDS
MORE THAN 800,000 FILES UPLOADED ON DROPBOX
NETFLIX
MORE THAN 87,000 HOURS OF VIDEO WATCHED
MORE THAN 5,500 CHECKINS ON FOURSQUARE
MORE THAN 25,000 POSTS ON TUMBLR
Eventbrite™
imgur
MORE THAN 2,000,000 MINUTES OF CALLS DONE BY SKYPE USERS
AROUND 200 EVENT TICKETS SOLD ON EVENTBRITE
MORE THAN 1000 IMAGES UPLOADED
MORE THAN 50 NEW REVIEWS
MORE THAN 500,000 APPS DOWNLOADED
MORE THAN 1,000,000 SWIPES
18,000 MATCHES ON TINDER
16,550 VIDEO VIEWS ON VIMEO
GO-Globe™
web design web applications identity seo

How to deal with data?

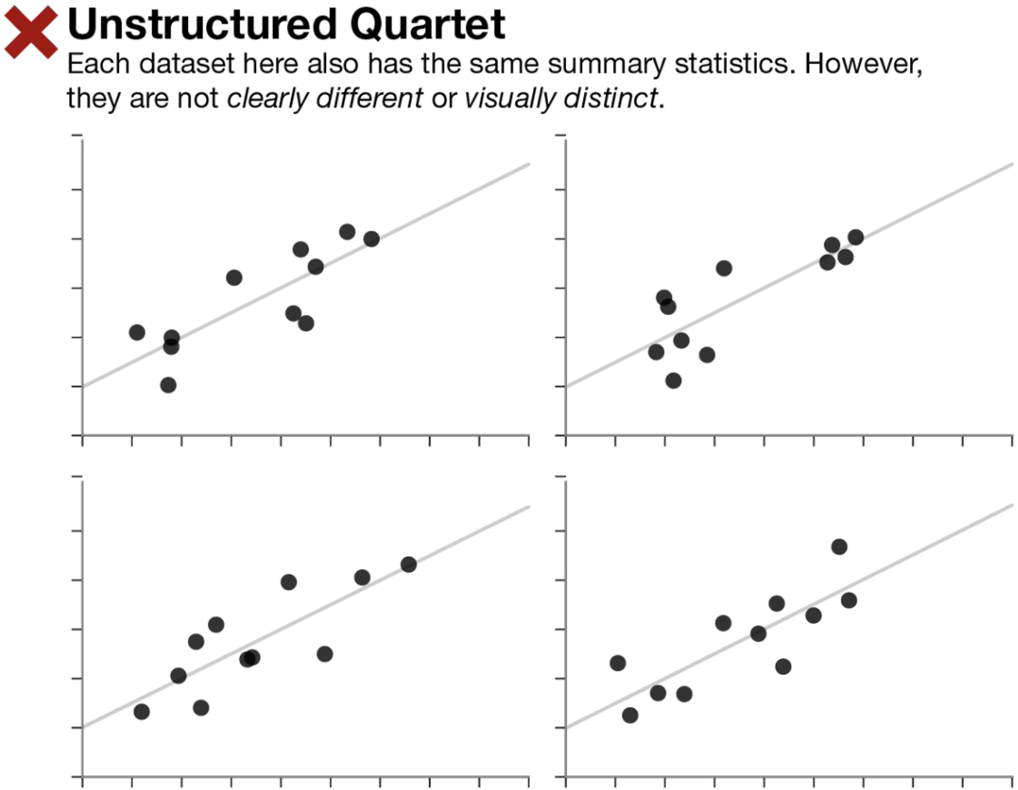
- **Observe the data**
- Summarize their ‘uncertainty’
- Search for constraints and invariances (prior knowledge)
- Find a model to explain the data

- Represent the data

- Find a solution to learn both the representation and the model for the data by exploiting advanced computing capabilities

- Avoid data bias

How to deal with data?



https://www.autodeskresearch.com/publications/samestats

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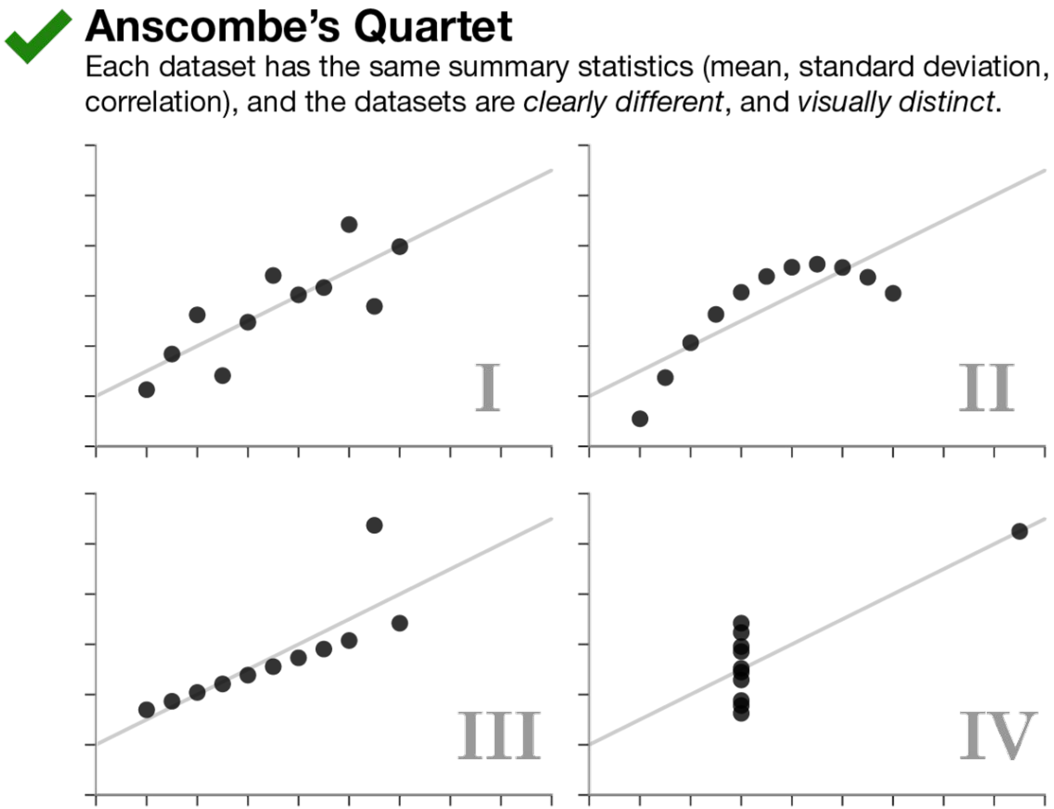

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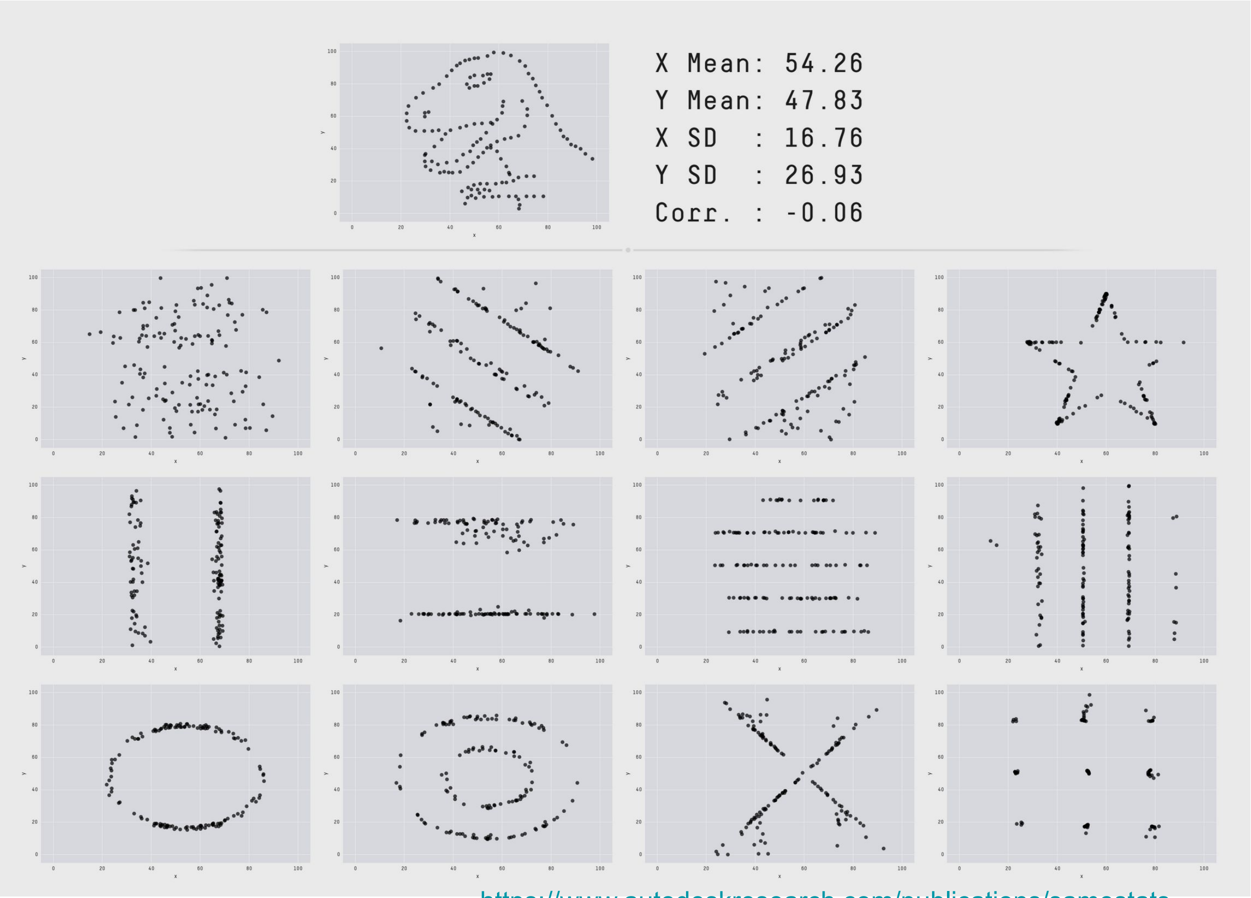
How to deal with data?

Datasaurus dataset

All these datasets have the same summary stats to 2 decimal places



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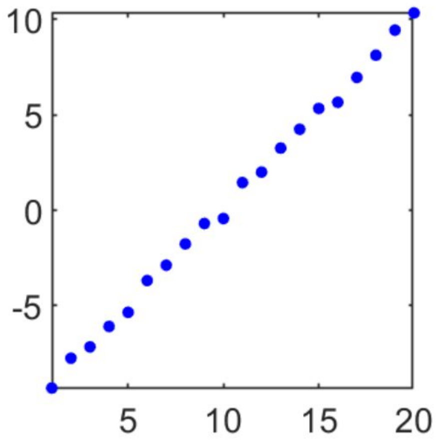
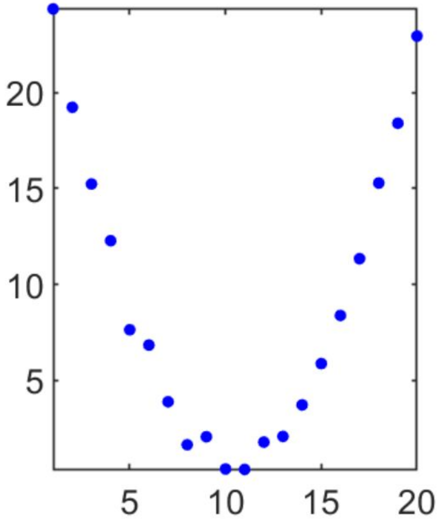
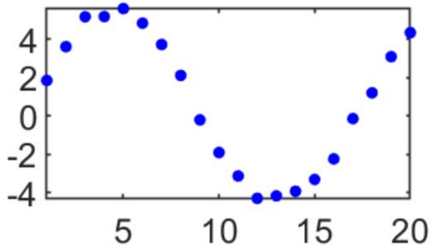
Models for describing the data

Main idea: Look at the world, identify what knowledge we have about it, and use this knowledge to construct our model.

The model is a map from input (x) to output (y)







slide credit: E Rodolà

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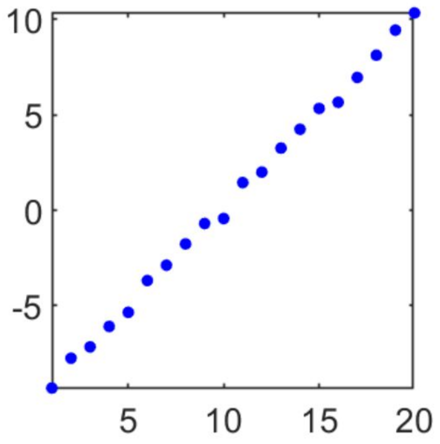
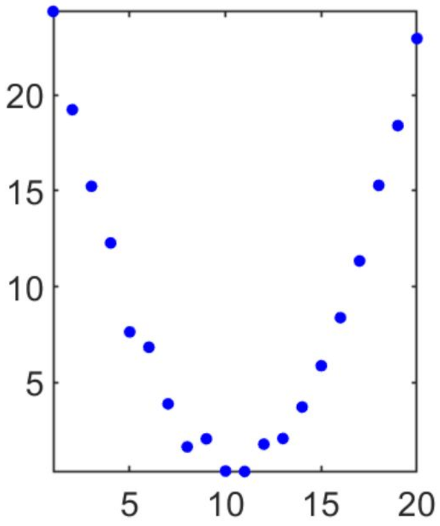
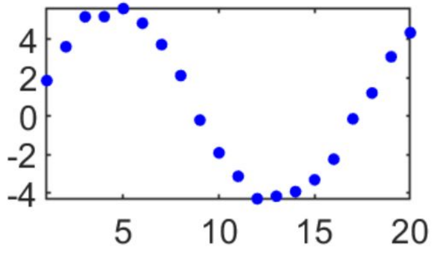


$$y = ax + b$$



$$y = ax^2 + bx + c$$



$$y = ax^3 + bx^2 + cx + d$$

slide credit: E Rodolà

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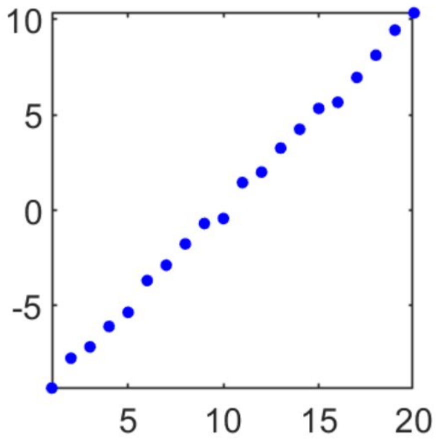

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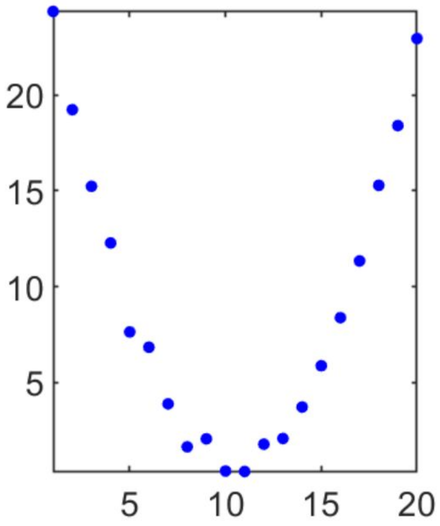

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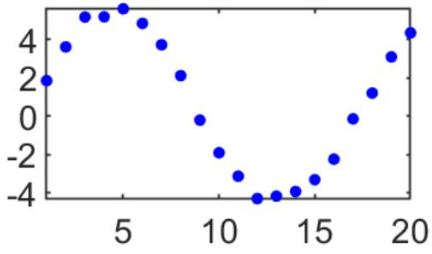


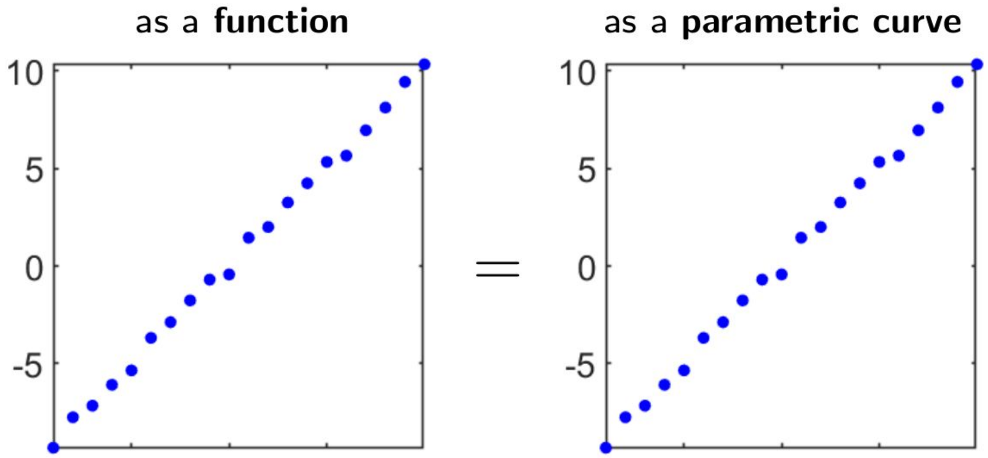
$y = a\sin(x) + bx + c$

AN ML

slide credit: E Rodolà

Choosing a Model Family

The same data can be described in different ways.
What is the “right” way?



$$y = ax + b$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} t + \begin{pmatrix} b_x \\ b_y \end{pmatrix}$$

slide credit: E Rodolà

Choosing a Model Family

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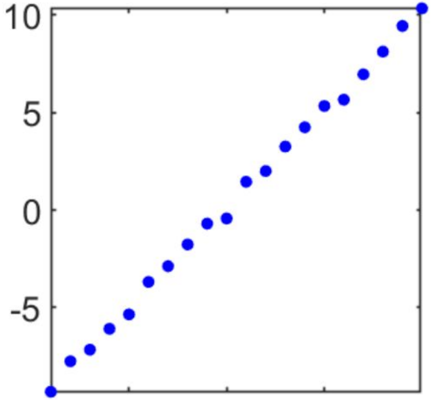
What is the “right” way?

as a **function**

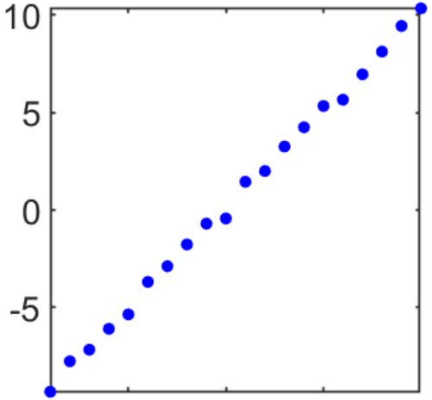


$$y = ax + b$$

2 weights

=

as a **parametric curve**



$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} t + \begin{pmatrix} b_x \\ b_y \end{pmatrix}$$

4 weights

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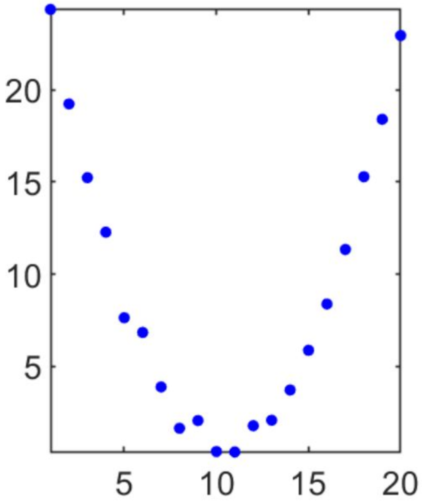

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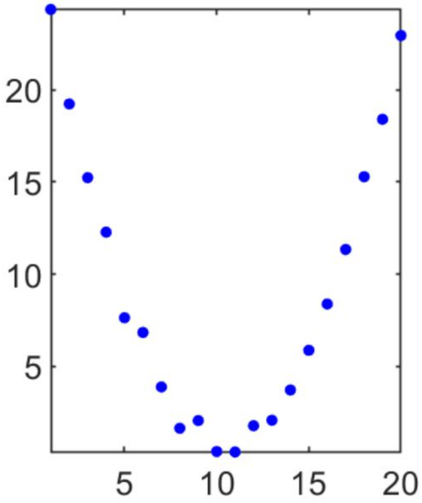

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} t^2 + \begin{pmatrix} b_x \\ b_y \end{pmatrix} t + \begin{pmatrix} c_x \\ c_y \end{pmatrix}$$

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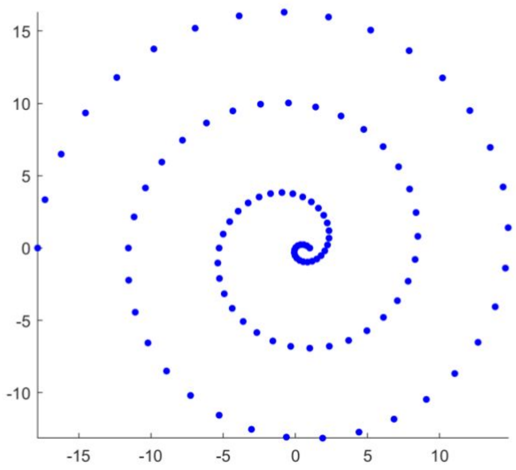
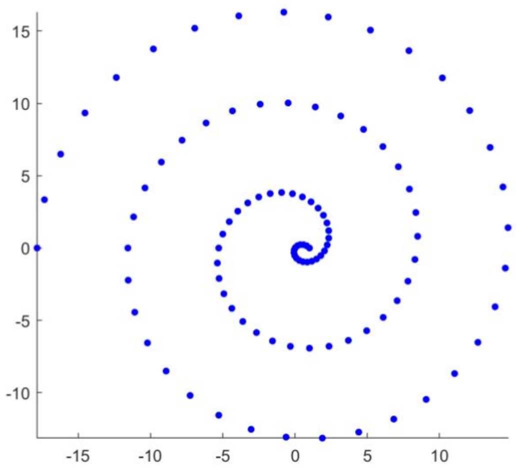


y is not a function of x

$=$



$$\begin{pmatrix} x \\ y \end{pmatrix} = a \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} (a - t)$$

slide credit: E Rodolà

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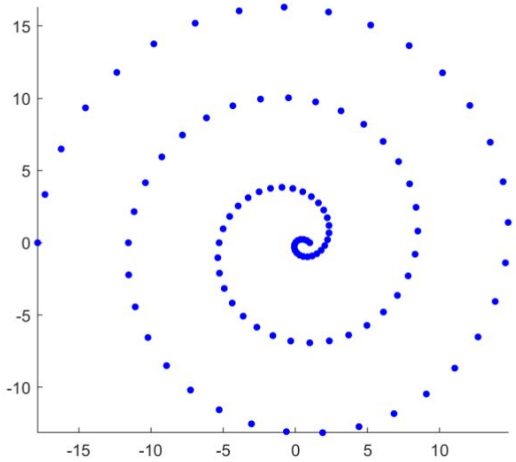
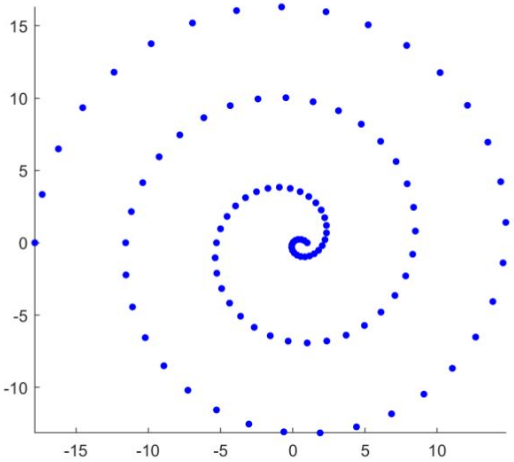


$$r = a\theta$$

(polar coordinates)

$=$

as a **parametric curve**



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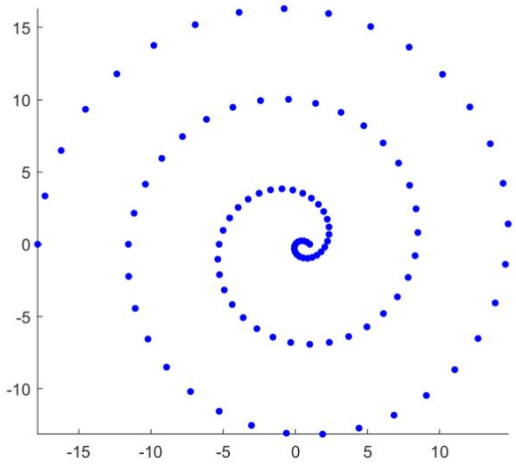


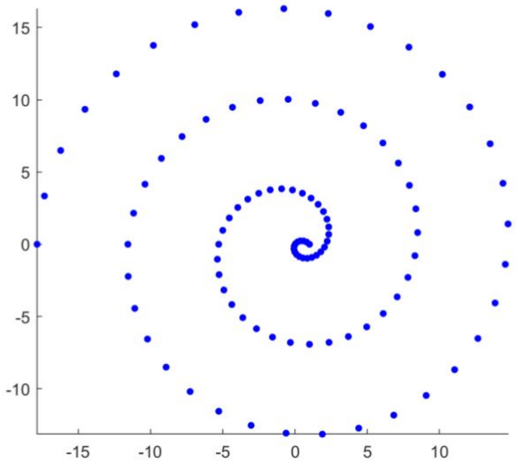
$$r = a\theta$$

(polar coordinates)

linear!

$=$

as a **parametric curve**



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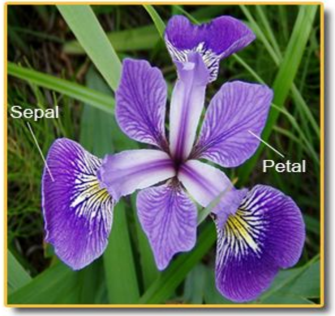
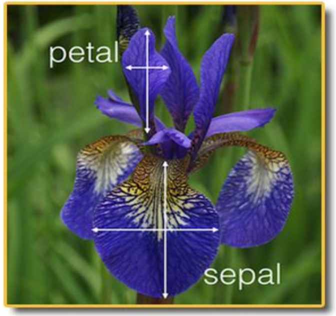
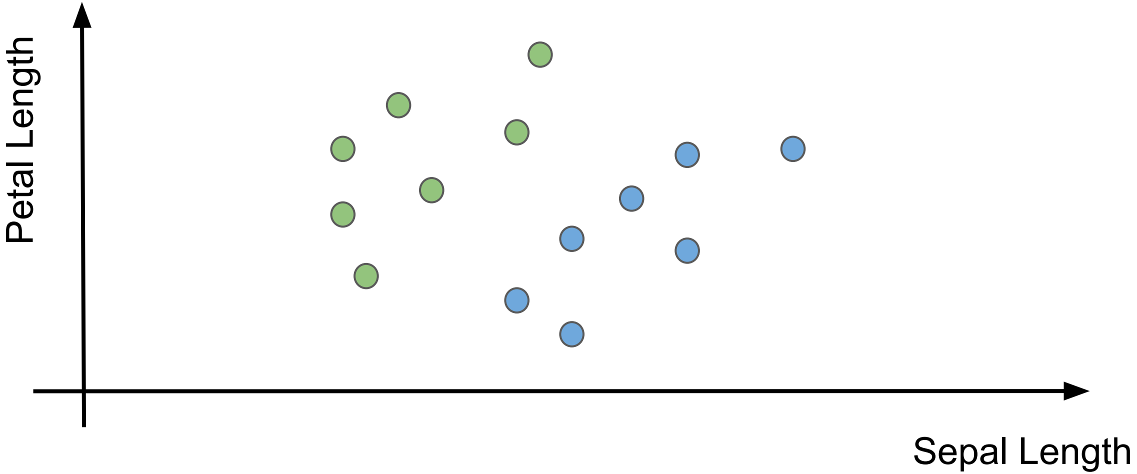
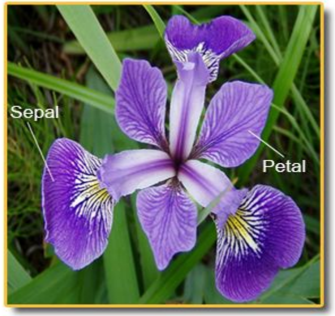


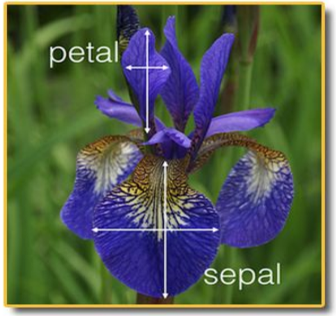




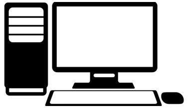
Representation Features

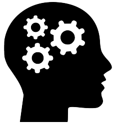
Choosing a Representation





Classification Model





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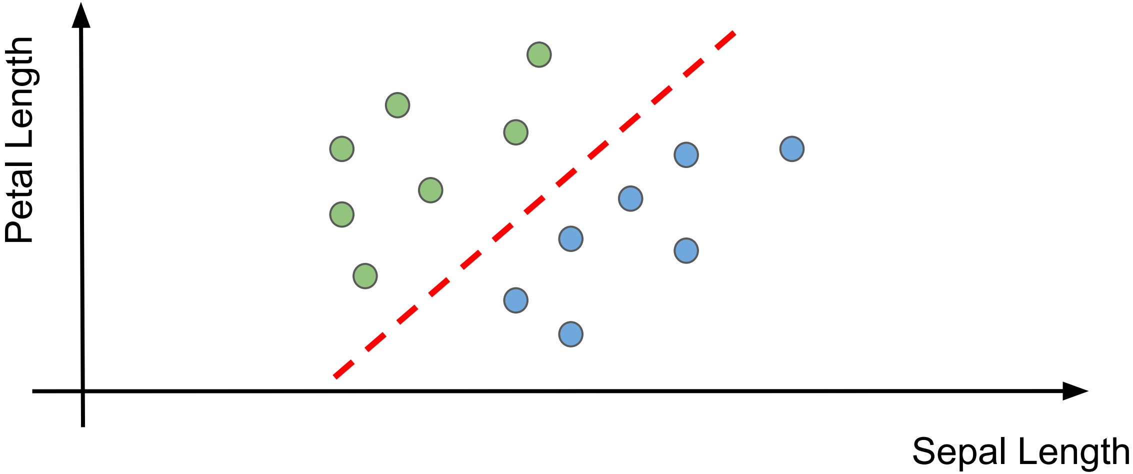

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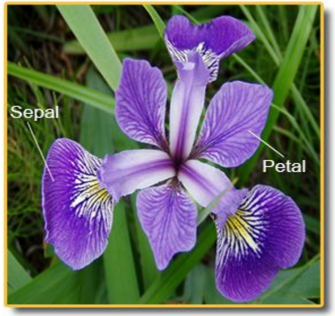

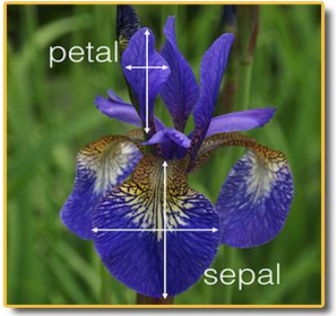

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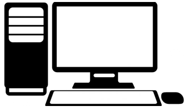

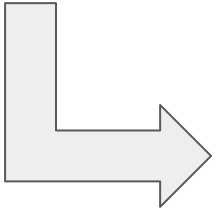

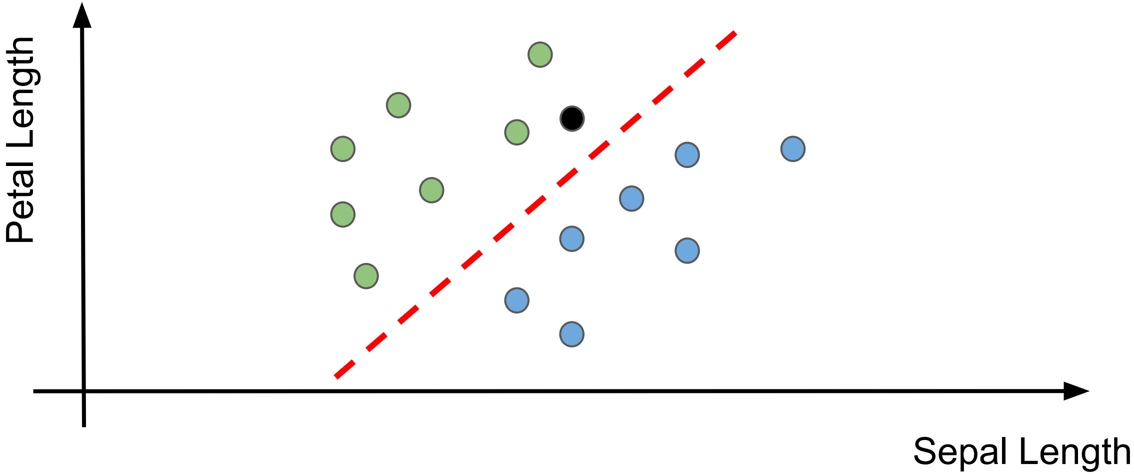

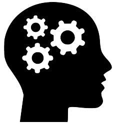

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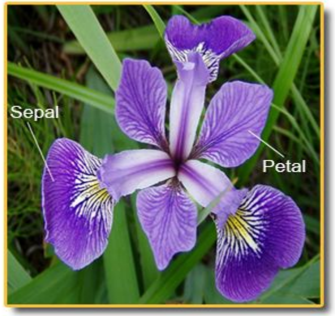

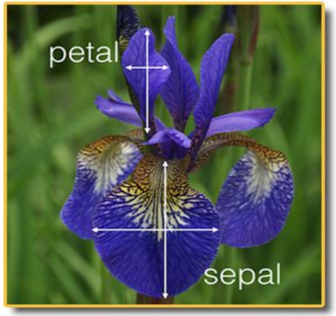

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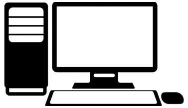

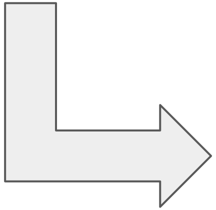

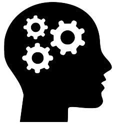

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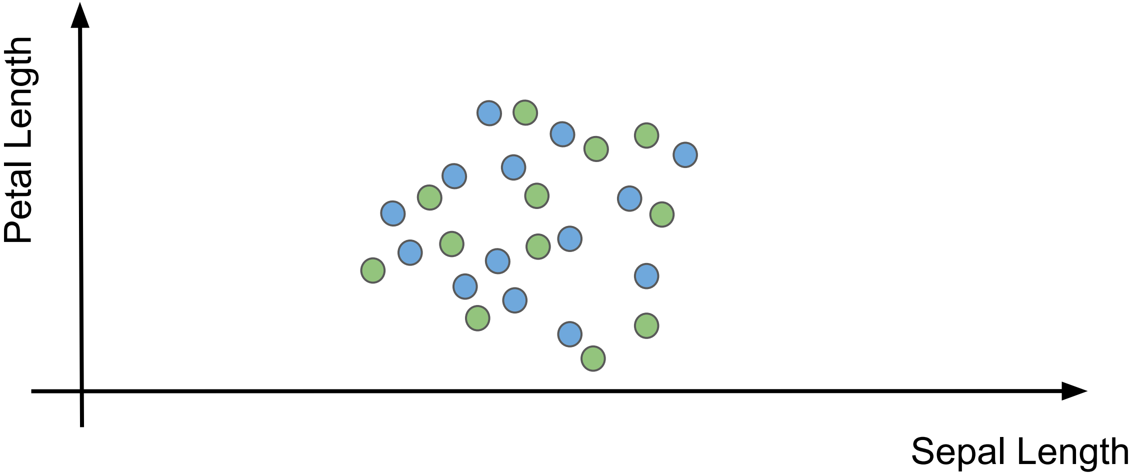

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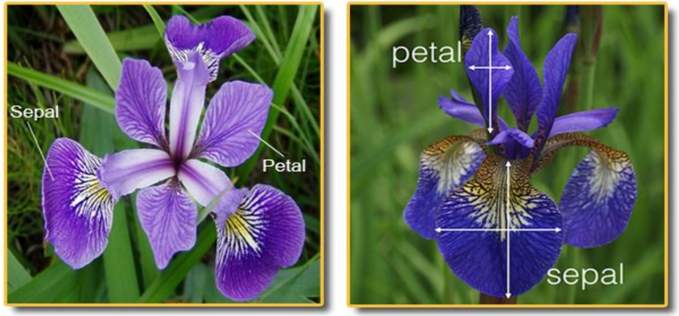
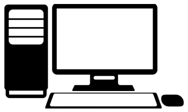
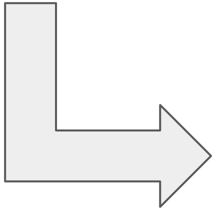
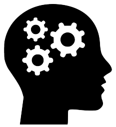


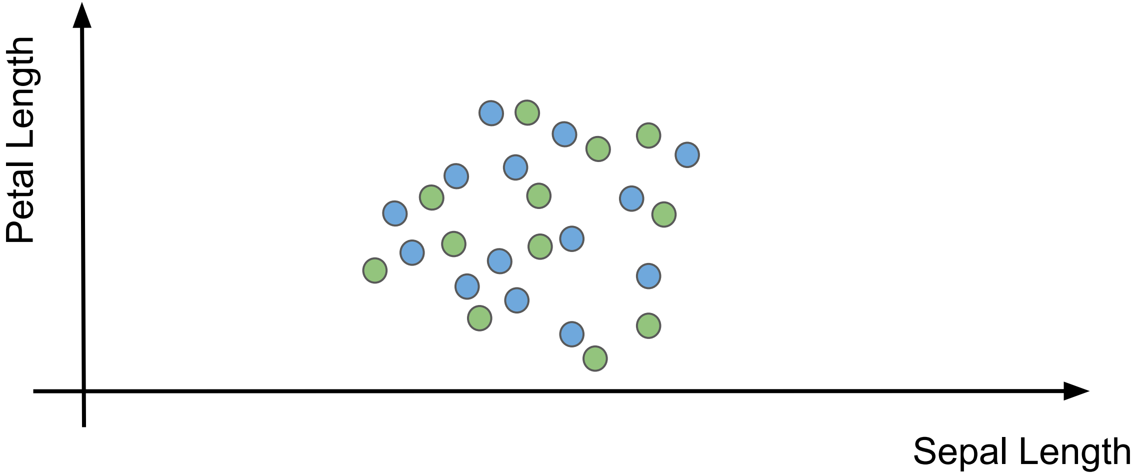
Classification
Model



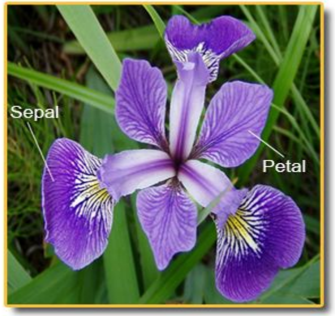


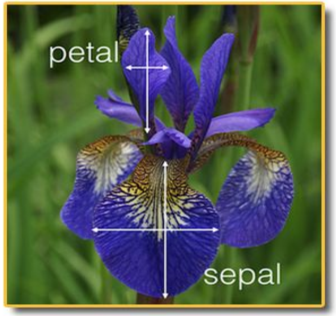


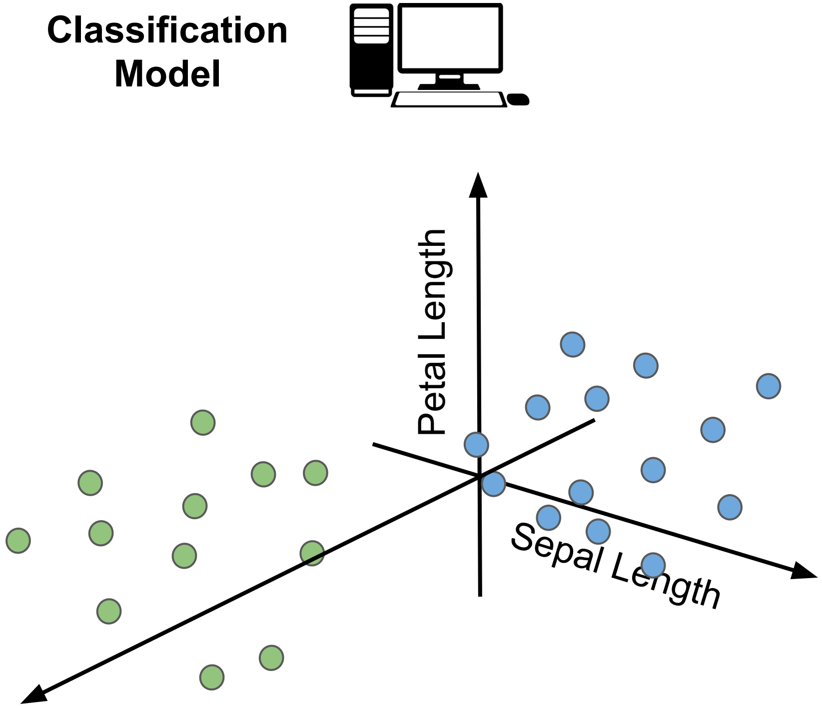
Representation
Features



Choosing a Representation

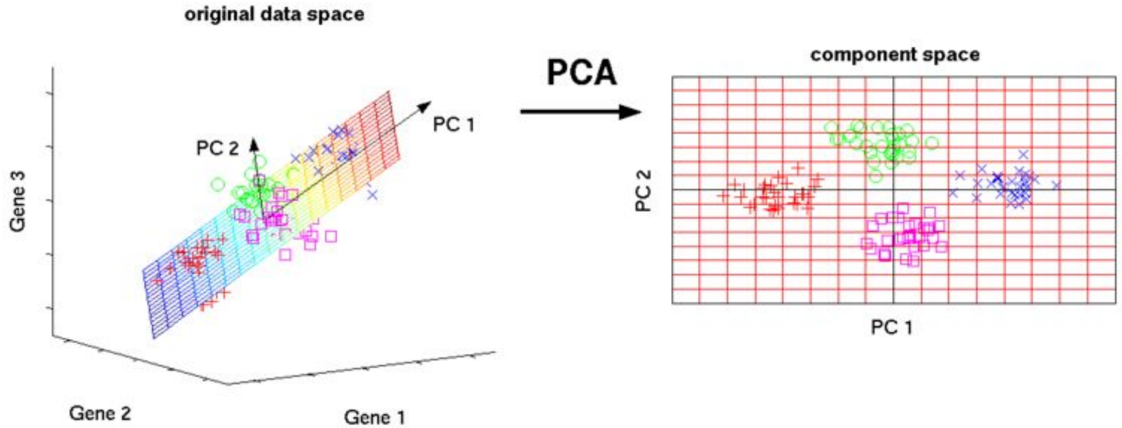


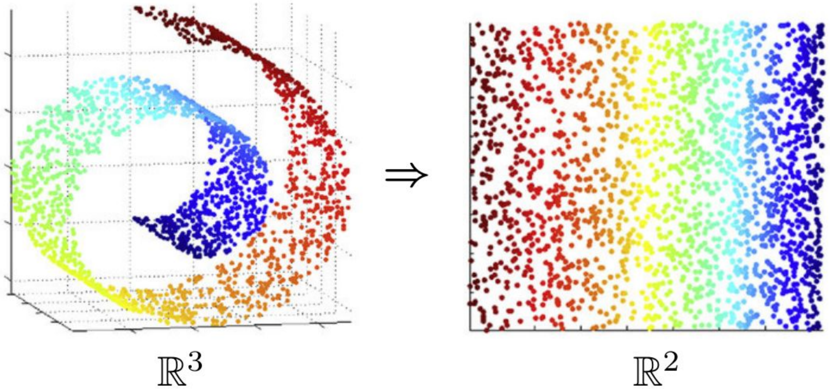




Choosing a Representation

Linear and Non-linear Dimensionality Reduction





How to deal with data?

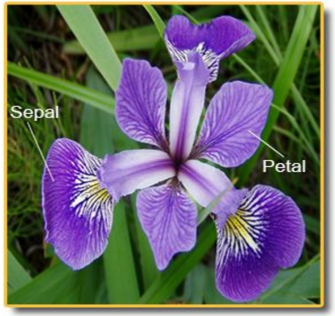
- Observe the data
- Summarize their ‘uncertainty’
- Search for constraints and invariances (prior knowledge)
- Find a model to explain the data

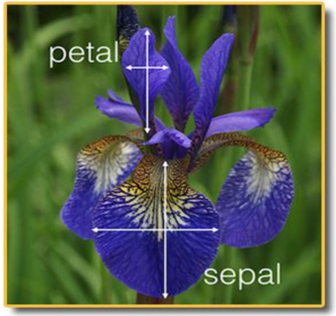
- Represent the data

- Find a solution to **learn both the representation and the model** for the data by exploiting **advanced computing capabilities**

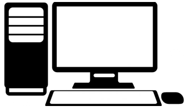
- Avoid data bias

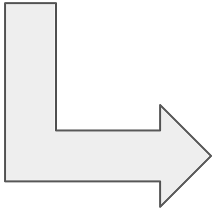
Deep Learning

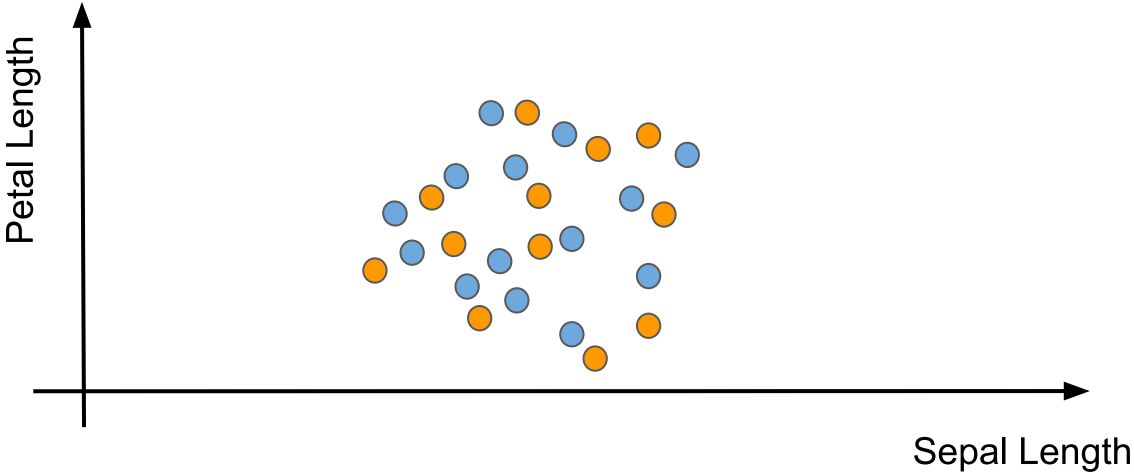


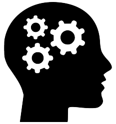


Classification Model



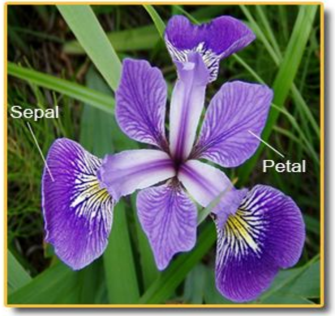


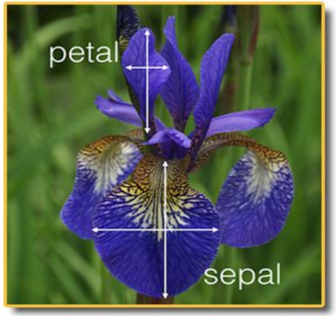




Representation Features

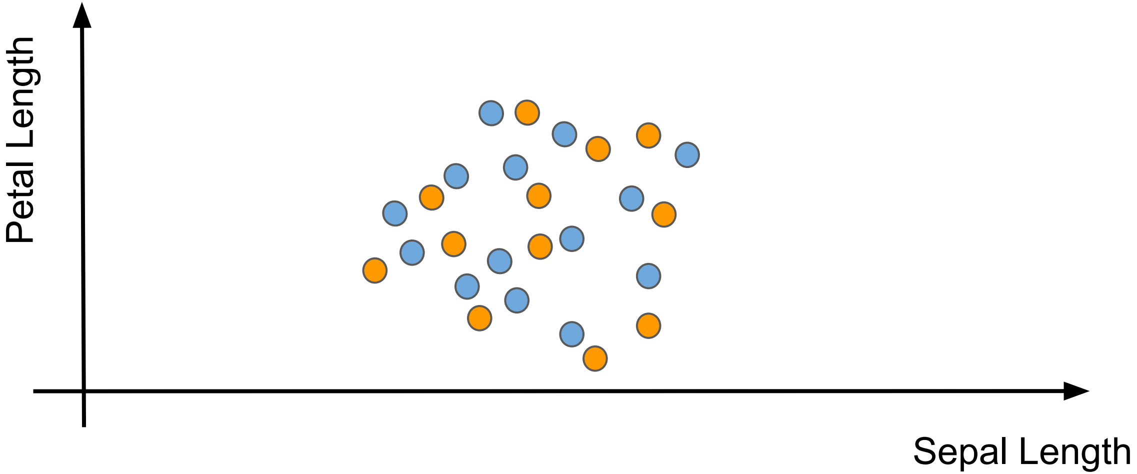
Deep Learning





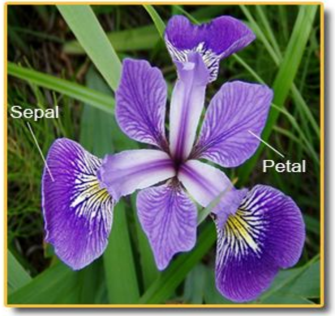
Classification Model

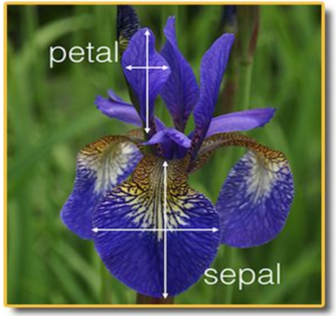
Representation Features

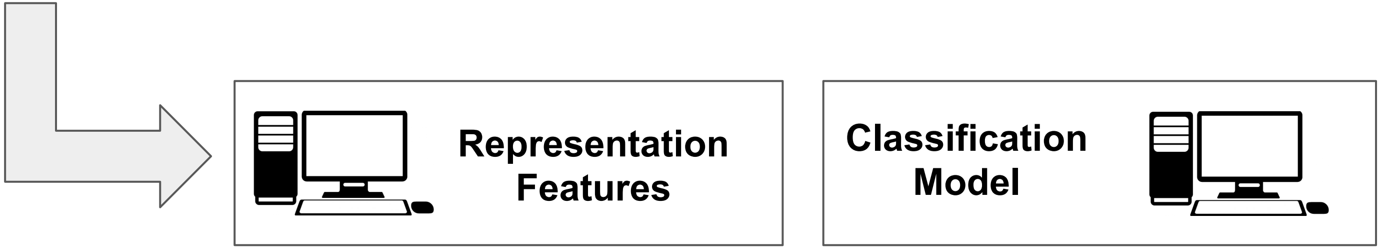


AN ML

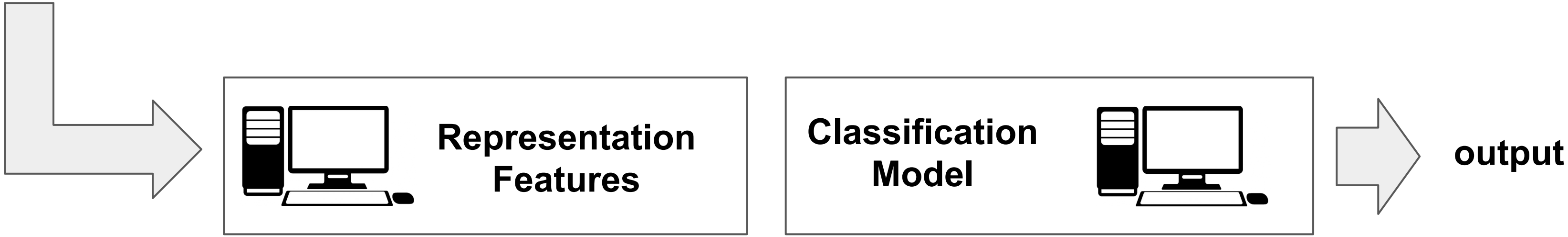
Deep Learning







Deep Learning



Deep Learning

Features are Task-driven

Speaking about features only makes sense if we are given a task to solve!

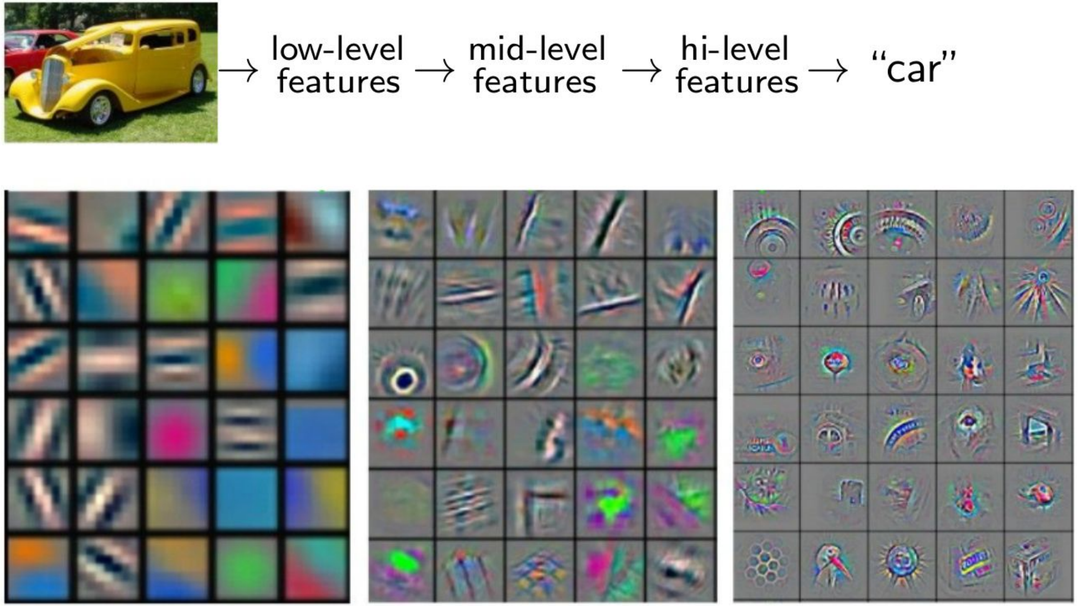
Is color important?

Rank, suit, and color are generic features, but specific problems determine what features are important for that task. What counts in spades, does not count in poker.

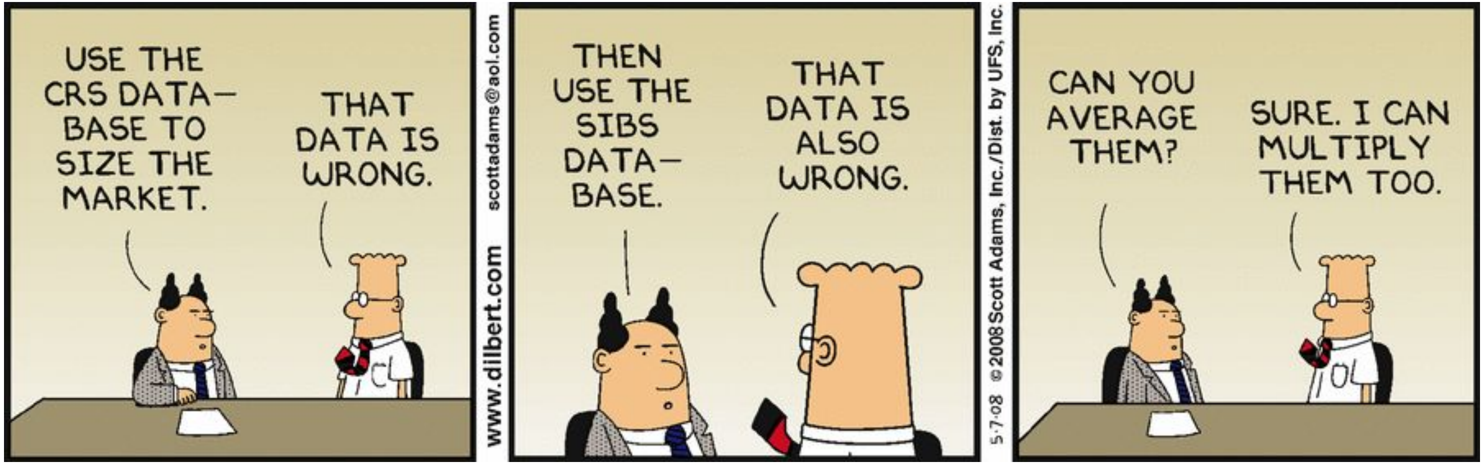
Deep Learning

Deep learning is a task-driven paradigm to extract patterns and latent features from given observations. However, features are not the focus of deep learning; rather, they are instrumental for the given task and drive the decision.

Example: Visual classification



Data Bias and Fairness



AI is objective only in the sense of learning what human teaches.

The data provided by human can be highly biased.

Data Bias and Fairness

	WHITE	AFRICAN AMERICAN
Labeled Higher Risk, But Didn't Re-Offend	23.5%	44.9%
Labeled Lower Risk, Yet Did Re-Offend	47.7%	28.0%

Overall, Northpointe's assessment tool correctly predicts recidivism 61 percent of the time. But blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend. It makes the opposite mistake among whites: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes. *(Source: ProPublica analysis of data from Broward County, Fla.)*

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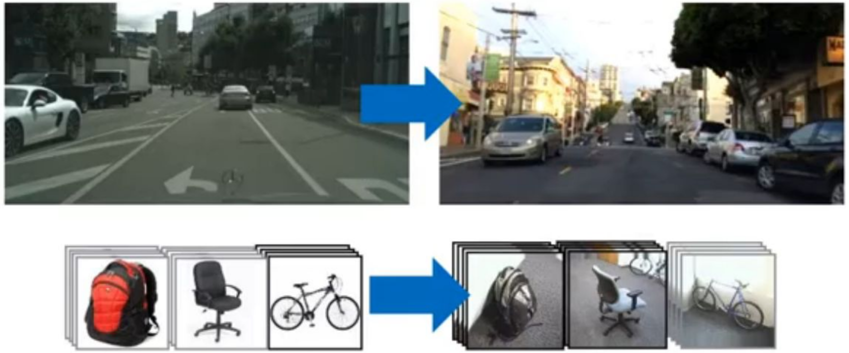
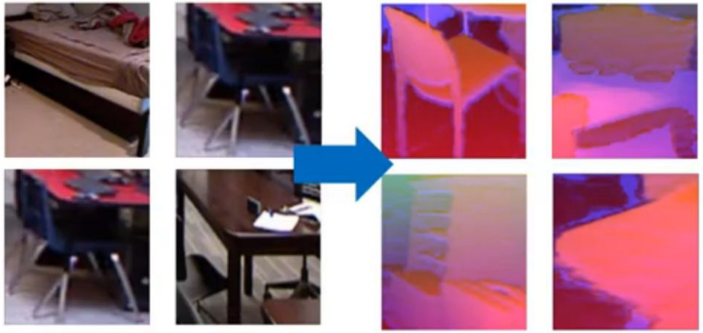
Data Bias - Adaptation needed

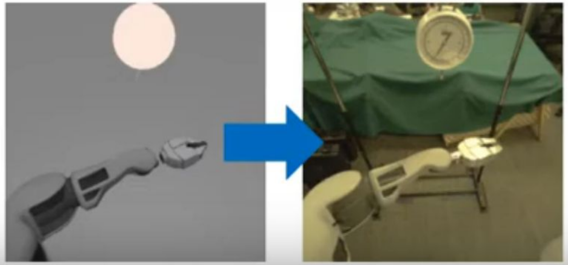
From dataset to dataset



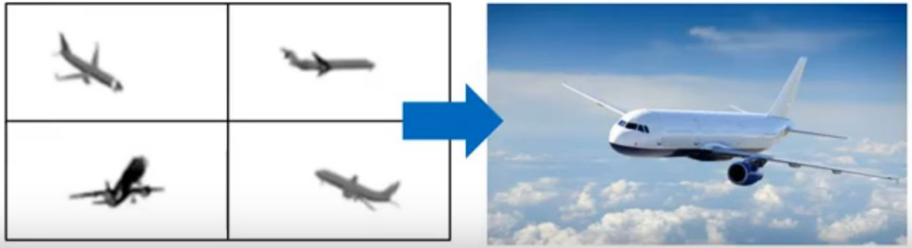
From RGB to depth



From simulated to real control



From CAD models to real images



slide credit: K. Saenko

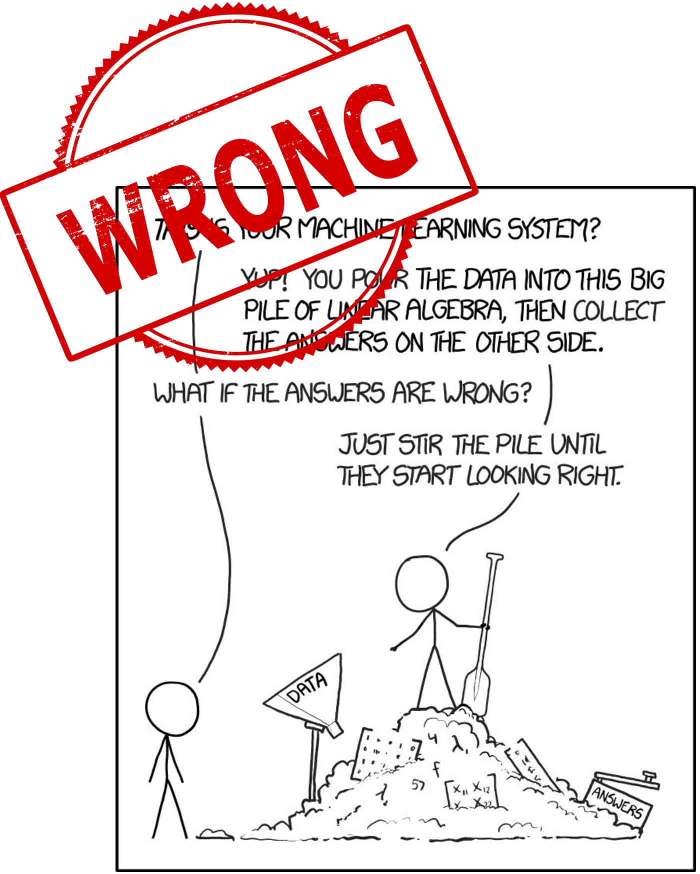
Summary

- Course and Exam presentation

- What is
  - artificial intelligence
  - machine learning
  - deep learning

- Learning: build a model

- How to deal with data
  - observe samples
  - feature representation
  - dataset bias and fairness



https://xkcd.com/1838/

<u>How to deal with data</u>

- Observe the data
- **Summarize their ‘uncertainty’**
- **Search for constraints and invariances (prior knowledge)**
- Find a model to explain the data
- Represent the data
- Find a solution to learn both the representation and the model for the data by exploiting advanced computing capabilities
- Avoid data bias

<u>Machine Learner Pseudo-Algorithm</u>

1. **Understand domain, prior knowledge, and goals**
2. Data integration, selection, cleaning, pre-processing, etc.
3. Learn models
4. Interpret results
5. Consolidate and deploy discovered knowledge
6. Go back to 1

Probabilistic reasoning